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The Emergence of the Dominant Design in Socio-Technical Transitions

A Case Study of Electric Aviation in Sweden

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Framväxten av den dominanta designen i sociotekniska övergångar

En fallstudie av elektrisk luftfart i Sverige

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Abstract

There is currently an ongoing transition into more sustainable aviation. One of the sustainable alternatives to the conventional fossil-fueled aircraft is electric aviation. The research aim of the study could be divided into two parts. The first part was to identify the barriers to the development of electric aviation in Sweden. The second part was to ascertain if and, in that case, how the process to reach dominant design can be affected by the actors. Empirical data was mainly gathered through semi-structured interviews. In order to address the research aim, a theoretical framework built upon theories from both multi-level perspective and dominant design theory was used. There were barriers arising from *Financial*, *Regulatory & Political*, *Business Model & Legitimacy*, and *Technical* conditions. The theoretical analysis implied the existence of two *socio-technical gates*. These gates consist of collaboration and legitimacy aspects. Metaphorically passing through these gates would make overcoming the relevant barriers much easier. This would also align all elements of the transition, which makes it possible for the actors to affect the process of achieving dominant design.

Keywords: electric aviation, socio-technical transitions, socio-technical gates, multi-level perspective, dominant design theory, legitimacy, actor collaboration, window of opportunity, dominant category.

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Sammanfattning

Det pågår en övergång till mer hållbart flyg. Ett exempel på dessa hållbara flyg är elektriska flygplan. Målet med studien kan delas upp i två delar. Den första delen är att identifiera barriärerna som hindrar utvecklingen av dessa elektriska flygplan. Den andra delen är att ta reda på om och i så fall hur processen för att nå en dominant design på marknaden kan påverkas av aktörerna. Insamlingen av data gjordes huvudsakligen med hjälp av semistrukturerade intervjuer. Teorier från "multi-level perspective" och dominant design-teori har använts. Resultaten visar på att det finns möjliga barriärer för utvecklingen inom områden av *ekonomi*, *regelverk*, *affärsmodeller och legitimitet* samt *teknik*. Analysen implicerar existensen av två *"socio-tekniska portar"*. Dessa består av legitimitet och samarbete mellan aktörer. Genom att bildligt passera genom dessa portar blir det mycket enklare att överbrygga alla relaterade barriärer. Detta riktar också alla element i övergången mot samma håll. Det i sin tur möjliggör att aktörerna kan påverka processen för att nå den dominanta designen.

Nyckelord: elektriska flygplan, socio-tekniska övergångar, socio-tekniska portar, multi-level perspective, dominant design-teori, legitimitet, kommunikation mellan aktörer, window of opportunity, dominant kategori.

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Foreword

This thesis was written during our Master's program in Industrial Management at KTH, the Royal Institute of Technology, in the spring of 2021. We would like to express our deepest appreciation to our supervisor during this time, Emrah Karakaya, who gave us invaluable advice and unwavering support throughout the duration. We also want to direct our appreciation toward our peer group and Frauke Urban, the examiner, for constructive feedback during the seminars to nudge us in the right direction. Lastly, we would like to thank all the aviation actors, especially those working with electric aviation, for participating in the interviews and providing us with critical insight and information.

1 Introduction

The introduction will cover the background of the study. This will be followed by the formulation of the thesis aim, research questions and delimitations.

1.1 Background

One of the prominent means of transportation in modern society is aviation. It is one of the quickest ways of connecting people and businesses all over the world. In the present, the utilized aircraft are to a great degree powered by fossil fuels. An end result of fossil fuel combustion is CO_2, which is emitted to the surrounding atmosphere during flights (Pawlak & Kuźniar, 2019). This alters the atmospheric greenhouse gas concentrations, which contribute to climate change. The aviation industry itself accounts for around two percent of the world's annual greenhouse gas emissions (Terrenoire et al., 2019). Furthermore, historical records show that air travel has an annual growth of 4-5%, which will create additional emissions (Barzkar & Ghassemi, 2020). The total projected greenhouse gas emissions are estimated to grow and reach up to 15% of the world's total greenhouse gas emissions in the upcoming years (Nava et al., 2018). Due to this concern, policy and decision-makers of the globe have established multiple directives in order to reverse the negative trend and reduce the impact of aviation (Erling, 2017).

This trend can also be found in Sweden, where the country has set ambitious goals to have zero net greenhouse gas emissions by the year 2045 (Karlsson et al., 2020). It is of great significance that the aviation industry thoroughly engages in this transition to enhance its sustainability. In 2017, the Swedish air travel accounted for 10 million tons of CO_2. The number of international trips made by each Swedish citizen has also roughly doubled over the last three decades, which further contributes to climate change (Kamb & Larsson, 2019).

A potential means for reducing the climate impact of the aviation industry is introducing electric aircraft. In recent studies, the potential of the full electric aircraft to minimize overall environmental impact has been assessed. Operating with battery and electric engine technology, full electric regional aircraft show great promise to minimize negative environmental impact, be greatly cost-competitive, as well as offering improved noise reduction (Viswanathan & Knapp, 2019). Full electric aircraft could reduce both nitrogen oxide and carbon dioxide emissions substantially, should strong progress in the technology be made (Schäfer et al., 2019). Schäfer et al. (2019) do, however, highlight that the first-generation full electric aircraft might not be as economically incentivized compared to the jet engine, in the current market conditions.

Another study showed that, even though the full electric aircraft have a greater energy consumption, the innovation has the potential to greatly lower CO_2 emissions, if and only if, the energy used is renewable (Gnadt et al., 2019). It is further supported by Sarlioglu and Morris (2015), that there are many plausible technological ways for electric aircraft to reduce emissions further. This implies that applying electric aircraft in commercial flights is more sustainable.

1.2 Formulation of the Thesis Problem

Electrified aviation is currently present in small pioneer projects for mostly private transports and recreational activities. It is yet to be introduced in larger contexts, such as commercial flights (Bauen

et al., 2020). Although the concept of electric aircraft is not new, the current state of the technology makes it infeasible for extensive commercial use. This is partly due to the lack of sufficient power in the electric aircraft for takeoff (Barzkar & Ghassemi, 2020). Besides the technical problems, the electric aircraft is expected to face socio-technical barriers upon its introduction (Han et al., 2019). Much research has been done to identify the technical obstacles and how these can be addressed. The same, however, is not true for the socio-technical aspects. In addition to this, it is not clear what the prevailing and viable design for commercial flights will be (Hoelzen et al., 2018; Chen et al., 2018; Barzkar & Ghassemi, 2020). Very few studies have been made on how to affect the process for reaching such designs (Yap & Truffer, 2019). Lastly, an introduction of electric aircraft into commercial flights would impact all the sector's actors and the entire socio-technical system as a whole; this will create a need for clarifying how to solve these obstacles (Gnadt et al., 2019). This is especially the case, since electric aircraft have not received much attention in literature. Its prevalence is, however, steadily increasing (Ratner et al., 2019).

1.3 The Research Questions

The study's aim is to understand how the process of reaching the dominant design for electric aviation in Sweden can be affected by the actors. The first step is to identify the socio-technical barriers for the development of the electric aircraft. Possible measures for these barriers will also be studied. The second step is to ascertain if, and in that case, how these measures can help the actors affect the process of reaching a dominant design for these electric aircraft. In this case, the dominant design refers to the electric aircraft model viable for commercial flights. Encapsulating this research aim are two research questions:

What are the current barriers for the development of electric aircraft in Sweden?

How can the actors affect the emergence of the dominant design in electric aircraft?

1.4 Delimitations

Sustainable aviation is a broad topic. To narrow down the scope, this thesis focused on electric aviation – on full electric aircraft in particular. The thesis was limited to Sweden. Also, given the novelty of the electric innovation, actors such as conventional fuel suppliers and conventional aircraft manufacturers have been excluded. It is to be further noted that the thesis was not only researching the technological aspects of the electric aircraft; prior projects, e.g., Solar Impulse 2, have already proven it possible to fly using electric aviation (Jaszczur et al., 2021). Instead, this thesis used a more holistic perspective that included the socio-technical aspects of the development, i.e., various socio-technical barriers. Another important factor that was deemed outside of the scope of the thesis was global pandemics, such as the Covid-19. While the virus has impacted the aviation industry, its future implications are hard to predict. Also, the focus was not on what the dominant category or dominant design could be, but rather the whole process itself.

The theoretical framework used in this thesis, consisted of only relevant parts from dominant design theory and the Multi-level Perspective. Additionally, only qualitative research methods were used. This means that this thesis was not predicting the future of the aviation industry based on empirical

patterns from other industries, but rather analyzing the immediate situation around the innovation and what this implied. This information primarily came from semi-structured interviews with people and organizations involved in the transition.

2 Literature review

Presented below are several theories and perspectives relevant to the research questions. Knowledge regarding the aviation industry, policies and the technical aspects of electric aircraft have also been gathered. This includes broad, structured and operational knowledge about the industry, current policies and sustainable initiatives in the aviation industry, and fundamental state-of-the-art information about different electric aircraft.

2.1 Transition Literature

2.1.1 The Multi-level Perspective

A very relevant perspective in socio-technical transitions is the multi-level perspective (MLP) (Geels & Schot, 2007; Geels, 2011; Geels et al., 2016). The articles introducing this perspective describe, among other things, the distinction between the socio-technical landscape, socio-technical regime and niche innovations and how these interact with each other, leading to new regimes (see *Figure 1*). Pressure from the changes at the landscape level and internal momentum from niche-innovations destabilizes the socio-technical regime. This then leads to transitions (Geels & Schot, 2007).

The basic premise of the MLP is that the development at three analytical levels results in transitions. These three levels are niches, socio-technical regimes, and the exogenous socio-technical landscape (Rip & Kemp, 1998; Geels, 2002). Exogenous refers to an external process of which the actors cannot affect. The niche-innovation level consists of market or technological niches. The socio-technical regime-level consists of actors and their activities that maintain linkages in socio-technical systems. The socio-technical landscape consists of elements such as cultural values or material arrangements (Geels, 2006). MLP has previously been used in socio-technical analysis of low-carbon transitions (Geels, 2012). Following the rise of concerns regarding environmental problems, the transport sector is faced with great challenges. Geels (2012) stated that greatly reducing carbon emissions in the transport sector can only be done by deep structural changes in the transport systems.

Geels (2011) conceptualized transitions in MLP as socio-technical systems with three analytical levels that consisted of heterogeneous configurations of elements. The elements in these socio-technical systems are maintained and changed by the various actors. Geels (2012) also stated that the MLP is broader than other approaches to sustainable development. Instead of focusing on a limited set of dimensions, the socio-technical approach instead highlights multi-level interactions between these critical elements of the transport systems. In the niche-analytic level, relevant experimental and demonstration projects were identified. These projects generally allow niche actors to learn about innovation in real-life circumstances. In the socio-technical regime-level, an important factor is the alignment of existing technologies, policies, regulations and infrastructure. Novelties must compete with technologies that benefit from these developed systems. The final analytical level, the socio-technical landscape, is the wider context. This influences both niche and regime dynamics. This includes physical structures, beliefs, media and trends.

According to Walrave et al. (2018), the MLP can also be used for strategic analysis – that is to identify the dynamics of important actors of the ecosystem. This may be especially beneficial in the areas of disruptive innovation or general path-breaking innovation. According to Christensen et al. (2013), disruptive innovation refers to the process of a smaller company with fewer resources

being able to successfully challenge established incumbent businesses. The disruptive process often begins in overlooked market segments such as low-end and or new-market footholds. Path-breaking innovation on the other hand is increasingly developed and commercialized by networks of co-creating actors. Similar to disruptive innovation, path-breaking innovation inherently challenges prevailing socio-technical regimes in a domain (Walrave et al., 2018).

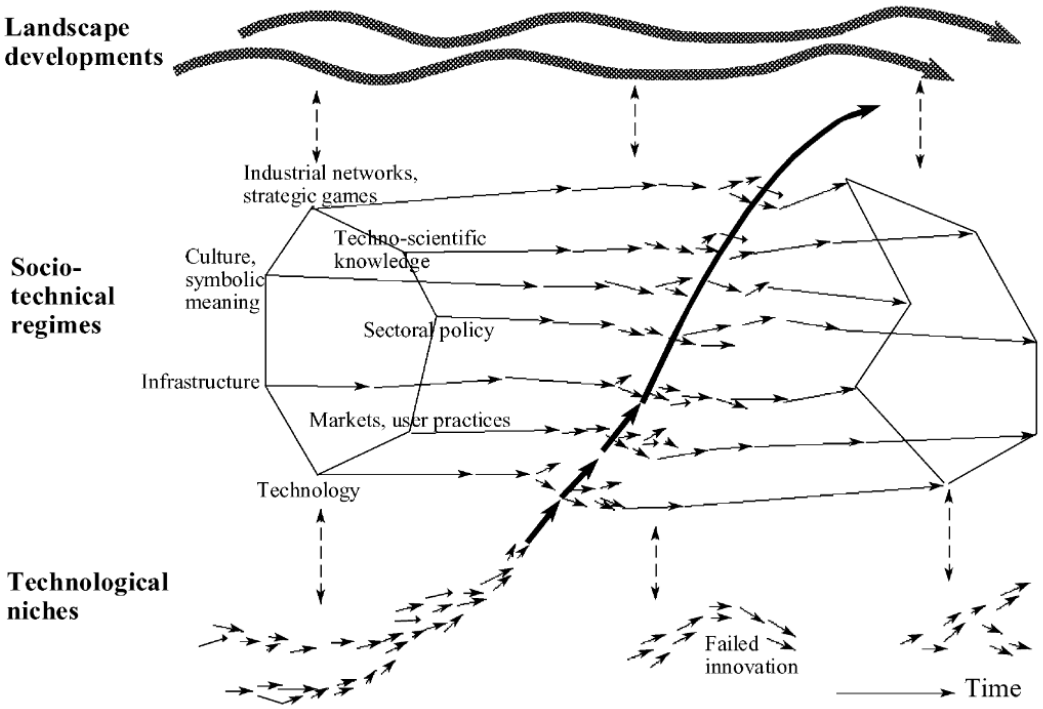


Figure 1: The three analytical levels of MLP by Geels (2002).

2.1.2 Dominant Design

In many product categories, technological evolution results in the emergence of a single product design that achieves market dominance (Srinivasan et al., 2006). For the emergence of this single product design – the dominant design, some key elements are timing, dominant category and the window of opportunity. The timing of entry into a new market is an important strategic choice for firms. The dominant category is a conceptual schema of the product itself. This is also what most stakeholders adhere to when referring to such products within the same market space. It is the abstract concept of the product itself. Lastly, the window of opportunity is a theoretical period of time where a firm entry starts with the convergence of the dominant category and ends with the emergence of the dominant design (Suarez et al., 2015).

Business environments in early stages of formation are named nascent markets. These are characterized by undefined or fleeting industry structures (Santos & Eisenhardt, 2009). Various socially constructed partitions or taxonomies that divide the social space into groupings of objects that are perceived to be similar, slowly emerge. These are known as categories. At these early stages of industries, differences in categorical positioning among different stakeholders, often reflect basic technological differences among products or basic understanding of the new industry. In the nascent market stage of the automobile, the same product and category were referred to as "velocipede", "locomobile", motorcycle" and so on (Kaplan & Tripsas, 2008; Suarez et al., 2015). This early stage is also known as categorical divergence, which is eventually followed by the convergence of the dominant category. This occurs when stakeholders' experience and familiarization increase, which leads to the abandonment of irrelevant dimensions along a natural preference of categories that gains traction over alternatives (Suarez et al., 2015). The concept of dominant category is related to the dominant design, but is in the end a socio-cognitive construct and not a product design (Bingham & Kahl, 2013).

The best timing of entry for firms is widely debated. There is no general consensus of what the best timing is. There are some people that speak for the first-mover advantages, such as resource preemption, customer switching costs and reputation advantages (Lieberman & Montgomery, 1988). A counterpoint would be that there are uncertainties in being first in an unknown market. One example would be the general awareness of such a market and also the demand for it. Investments in a technological path that may or may not be inconsistent with the dominant design is disastrous for many firms (Bohlmann, Golder, & Mitra, 2002; Utterbeck & Acee, 2005; Suarez et al., 2015). Instead, most research has left entry-timing advantages and moved on to the window of opportunity. What marks the end of these windows is clear – this being the emergence of the dominant design, which is also confirmed by Anderson and Tushman (1990). However, what marks the beginning is still unclear. Suarez et al. (2015) are of the opinion that the concept of the dominant category offers the necessary explanation to what commences the window of opportunity. Suarez's hypothesis is that the convergence of the dominant category marks the beginning of the window of opportunity, as much of prior uncertainty dissipates.

The usage of the window of opportunity varies for different firms. Let us take sustainable development as an example. Policy makers in countries with industrial transitions are confronted with two challenges. These are how to implement cleaner technologies and infrastructures while also promoting rapid industrial development. The trade-off can be interpreted as providing windows of opportunity for gaining strong dominance in these new cleantech industries (Yap & Truffer, 2019). Analyzing these windows of opportunities is a prevailing theme in any strategic positioning

in the transitions. Here, it is common to perceive these windows of opportunities as exogenous events that firms need to respond to (Lee & Malerba, 2017). Ferràs-Hernández et al. (2017) present similar ideas. After researching the disruption of the automotive industry, it was concluded that disruptive innovations were led by outsiders with entrepreneurial experience. According to them, these innovations would potentially lead to new dominant designs. This essentially leads to firms adopting strategies for predicting and reacting to these outsider events.

There are theories that state that it is possible to turn these windows of opportunity to insider events that firms are capable of influencing. Yap and Truffer (2019) state that the windows of opportunity can be made endogenous, i.e., an internal and controllable process, through the interplay of different actors trying to shape different layers of the selection environment of the sector. This interplays with Walrave et al. (2018) theories about the MLP being a pertinent tool for strategic analysis in such conditions. It should be possible to identify the key actors that may influence the window of opportunity, which consequently will also influence the emergence of the dominant design. The same key actors form a network which will essentially determine whether the innovation will achieve dominant design or not.

2.1.3 Related Work

There is a lack of literature applying socio-technical transition theory of the field of aviation. However, there are still a scarce number of articles that have applied theories such as MLP on the field. Geels (2006) wrote about the co-evolution and multi-level dynamics in transitions of the aviation industry, specifically the shift from propeller to jet-engine. Different co-evolutionary approaches address different levels and units of analysis. The main distinction should be made between the level of emerging innovations, the level of existing configurations, and the macro-level of the society as a whole. Through the research, it was proven that transitions in socio-technical systems can be studied not only as technological discontinuities, but also long-run co-evolutionary processes. With similarities to Smith et al. (2010), Geels states that the transitions come about, when the dynamics of all three levels of MLP link up and reinforce each other. The actors that drive the change involve not only firms, but also the market, infrastructure, user groups, culture, and regulations.

Similarly, Spinardi and Slayton (2015) used the MLP to gain a deeper understanding of environmental transitions. Their article analyzed the transition into green aviation. Differentiating itself from other works in the field, the article refined the MLP. Rather than predefining particular innovations as either radical, conservative, mature, or immature, they instead used the interpretation of how actors conceive such terms. An innovation that may seem immature to one actor, might appear as mature to another. The conclusion was that MLP could provide insight in how the niche, regime, and landscape dynamics allows for transitions. In their case, this was how the turbofan displaced the propeller-pistons in long-range aircraft.

2.2 Aviation Literature

2.2.1 The Aviation Industry

Aviation allows for a unique means of transportation through the air. The large and accessible global air transport network spans continents, massively expanding local access to international supplies and markets, facilitating cultural and social exchange, and improving emergency and humanitarian

response capabilities during disasters and public health emergencies (Große et al., 2019). The aviation industry is dependent on its connected components for enabling transportation. These components include air transport control centers (ATCC), airlines, airports, aircraft production, fuel suppliers, and other transport facilities that are closely related to the aviation industry (Dožić, 2019).

In Sweden, the airports are either owned by private actors, municipalities or the state (Marklund, 2019). According to Große et al. (2019) the total of 248 airports in Sweden can be divided into the types: *National*, *Regional* or *"Other"* airports. The number of annual passengers in the national airports amounts to 750,000. This number of passengers is the criteria for an airport to fall under the category of national airports. The "other airports" are airports within a two-hour distance to Arlanda airport by car. Regional airports are the airports that do not fulfill either of these criteria. Of these 248 airports, approximately 15% manage commercial flights in the form of regular flight or charter lines (Trafikanalys, 2020). Swedavia and Svenska regionala flygplatser (SRF) are two major entities that manage some of these airports, Swedavia being supported by the government and SRF being dependent on aviation fees (Westin, 2020). Swedavia holds, through their airports, the most passengers for both domestic and international flights. For domestic flights in Sweden, Scandinavian Airlines (SAS) is the largest of the airlines, followed by Braathens Regional Aviation AB and Norweigan Airlines.

The total number of passengers with a Swedish citizenship has increased substantially in comparison with the national traveling levels of the 1990s. This has resulted in CO_2 emissions reaching new levels. In 2014, the CO_2 emissions reached 11 million tons (Kamb et al., 2016). The number of passengers reached approximately 39 million in 2018. The international flights covered 30 million passengers alone, while the national flights covered the remaining 9 million passengers (Große et al., 2019). In 2014, 50% of the total Swedish GDP stemmed from exports. This percentage is above the European Union's average and made Sweden the country most dependent on export throughout the whole EU. Furthermore, 20% of Swedish foreign trade is accomplished by aviation. It is thus essential that the industry operates with efficiency and is secure. The Swedish aviation industry contributes to the national GDP. Its share corresponded to 1.7% of the total annual Swedish GDP, which is equivalent to 53 million SEK. Furthermore, the industry provided 9 million SEK, in the form of tax revenues, to the nation's state budget. The Swedish aviation industry provides over 80,000 job opportunities. These jobs include among other things, both engineers as well as service staff (Svenskt Flyg, 2014).

Not all aviation is financially profitable in Sweden. A portion of the domestic flights has been established by Trafikverket to provide adequate accessibility to larger airports, e.g., Stockholm-Arlanda, for the citizens. This is under the European Parliament and Council Enactment number 1008/2008, written September 24th 2008 (henceforth, referred to as the *EU-aviation enactment*). These flights follow specific routes, established by both governmental and private actors. The largest provider of ATCCs, which guides the ongoing flights within the country, is Luftfartsverket (LFV). Some of the responsibility and tasks have, however, been delegated to new actors such as Swedavia and ACR (Große et al., 2019; Trafikanalys, 2020).

Since the Covid-19 pandemic, however, the aviation trends have taken a drastic decline. This is due to the regulations of the worldwide quarantine to lower the risk of infections (Zhang et al., 2020). It was predicted that the global airline industry would lose around 250 billion US dollars in revenue by

2020. This is due to the fact that the majority of the countries have agreed to close their borders to foreign travels (Salman et al., 2020). In addition to the pandemic, the Swedish citizens have begun to fly less in general. This is due to the emerging awareness of the climate impact. The citizens have begun to refrain from flying and have chosen to use more sustainable options, such as trains. This trend of refraining from flying can be defined as the phenomena of *"Flygskam"* (henceforth referred to as *"flight-shame"*, i.e., the shame of flying), which is increasing not only in Sweden but all around the globe (Söderberg & Wormbs, 2019).

2.2.2 Aviation Policies and Directives

Global aviation contributes to climate change. This is through emissions of greenhouse gases, e.g., carbon dioxide CO_2 and nitrogen oxides NO_x (Lee et al., 2021). A majority of the countries in the world have, in response, set targets for sustainability in the Paris Agreement (Lee, 2018). The goal was to reduce greenhouse gas emission to a zero-net level or possibly a negative output level. Policy makers are currently impacting the aviation industry by directing actors to align with sustainable goals. This can be done by incorporating new sustainable innovations, e.g., alternative fuels (Nair & Paulose, 2014). All actors are, however, not wholeheartedly compliant with the emerging sustainable policies and regulations. Some actors will actively resist such implementation due to detrimental effects on their operating businesses (Peeters et al., 2016).

The EU has laid out a strategy program for the aviation industry. One of the cornerstones of this program is to unleash the full potential of innovations, new technologies and investments of the aviation industry (Schulte-Strathaus, 2017). In addition to this, the EU has put in place a zero net emission goal, i.e., the Green Deal, which is to be fulfilled until the year 2050. According to the *Green Deal*, circular economy should be adopted and lost biodiversity should be restored in the union's sectors that impact the climate. The most important sectors affected are private and public transportation. This includes the aviation industry. To speed up the process, the EU will provide economical and technical support to the aviation industry. Furthermore, the European aviation actors have taken their own measures for sustainability in the *Carbon Offsetting and Reduction Scheme for International Aviation* (CORSIA) initiative (Eisenhut et al., 2021).

Sweden aims to be a leading actor for sustainability, within the committed countries of the EU. Therefore, the country has begun implementing new initiatives to become more sustainable. In 2017, the Swedish government established a new climate framework, consisting of a climate- law, goal and council. The intended long-term goal is to have a zero-net level of greenhouse gas emissions by the year 2045 (Lövin, 2017). The climate framework has impacted governmental transport agencies such as Trafikverket, Transportstyrelsen and Trafikanalys. The agencies are directed to report and follow-up on efforts on their policies for a sustainable transportation system in Sweden. Several of these directives have already been implemented in the agencies from prior objectives. The operations of said agencies have therefore not changed significantly (Berndt, 2018).

In the Swedish aviation industry, an initiative called *"Fossilfritt flyg 2045"* was launched by the actors SAS, Swedavia and RISE. The initiative was directed and financed by Energimyndigheten and was a two-year project, from 2019 to 2020. The goal for the initiative was to create a national platform to bring together actors that were engaged in the development of innovations within the aviation industry, e.g., electric aviation projects. The sectoral goals were first to have all domestic

flights fossil-free by 2030 and secondly to have all nationally departing flights be fossil-free by 2045 (Al-Ghussein & Talalasova, 2021).

2.2.3 Electric Aircraft Technology

The conventional type of aircraft are fueled by petroleum based fuels distilled from crude oil, one of the more prominent ones being *Jet A* (Wei et al., 2019). Due to raised concern about conventional fuels' negative impact on the global atmosphere, new alternative fuels have begun to be created and developed. These include biofuels, hydrogen and electrification. Electric aircraft have lately received a great deal of attention; due to its potential for commercial flight. It has also been adopted in small areas (Bauen et al., 2020). One example of an emerging electric aircraft is *Eviation Alice*. It was claimed to be certified and commercially viable as early as 2024 (Baumeister et al., 2020; Eviation, 2021). Another example is the American company Wright Electric, which is collaborating with Easy-Jet to develop a fully electric aircraft with a 186-passenger capacity and flight range of 500 kms. The aircraft is set to be fully operational in 2030 and is considered for the flight route between London and Amsterdam (Trafikanalys, 2020). In Sweden, an example is the company Heart Aerospace that has recently started to develop an electric aircraft – the Heart Aerospace ES-19 (henceforth, referenced as ES-19). It is expected to have a flight range of 400 kms. The aircraft is aimed to be commercially viable and certified for continuous production in 2026 (Baumeister et al., 2020; Heart Aerospace, 2021).

All aircraft that are to be introduced to the market have to go through the European Union Aviation Safety Agency (EASA). This is also the case for all electric aircraft. The agency evaluates and certifies the aircraft for airworthiness and develops common standards for the aviation industry. This is in terms of aircraft design, manufacturing and maintenance. Moreover, the agency has educational processes for flight crew members, air traffic managers and air navigators. Its main mission consists of establishing global security in aviation for all EU members. The mission stands upon four pillars which allows the agency to pursue its mission. One of these pillars establishes EASA as the sole agency with the authority to issue valid certification of aircraft and its components in the EU (Galván, 2020).

The dominant category of electric aviation is not yet fully determined. In the present, there are two major types of possible configurations. As illustrated by *Figure 2*, this includes the battery-powered and hydrogen fuel cell configuration (Bauen et al., 2020). Within the battery-powered configuration, there are hybrid electric aircraft (HEA) and full electric aircraft (FEA). Hydrogen fuel cell technology is currently also categorized as an electric aircraft. Hydrogen has been a topic of interest for a long time and it has been considered a promising solution in the future (Verstraete, 2013; Baroutaji et al., 2019).

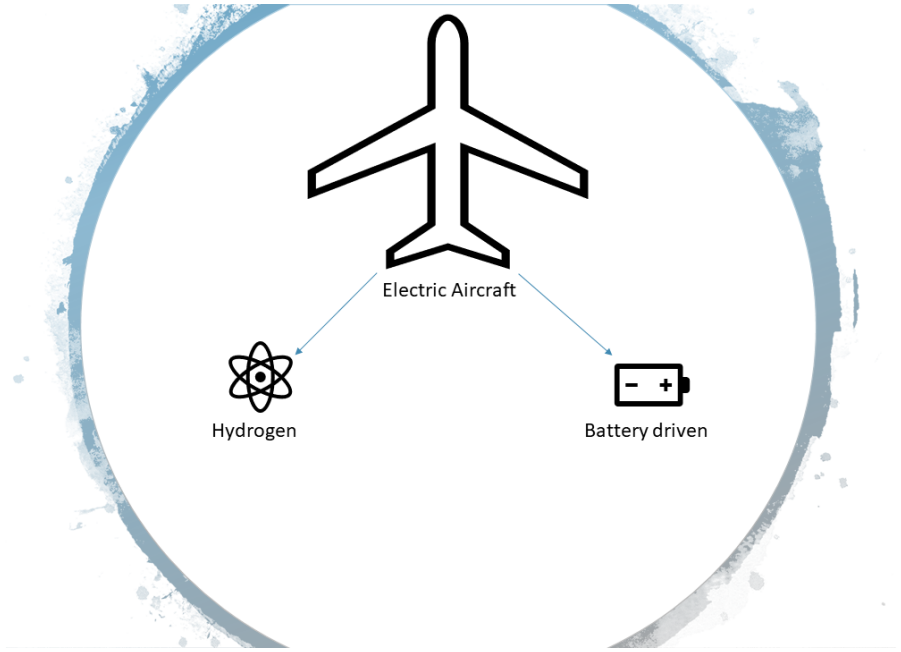


Figure 2: Illustration of the two technological configurations of electric aircraft.

The HEA configuration can be seen as a further development of the conventional aircraft, where an electric aspect is introduced to the jet engine system. The idea is to increase the propulsion efficiency with the help of an electric component in the overall conventional aircraft design, as seen in *Figure 3*. In recent developments, it is possible for HEA to reach a full hybrid configuration. This is considered viable for some actors. The hybrid configuration provides the aircraft with additional power and simultaneously reduces emissions during specific moments in the flying, such as takeoff or landing (Schäfer et al., 2019). Furthermore, full electric aircraft still need further development to be adequate for more extensive commercial flights, whereas the hybrid counterpart is already fully operational (Wheeler, 2016).

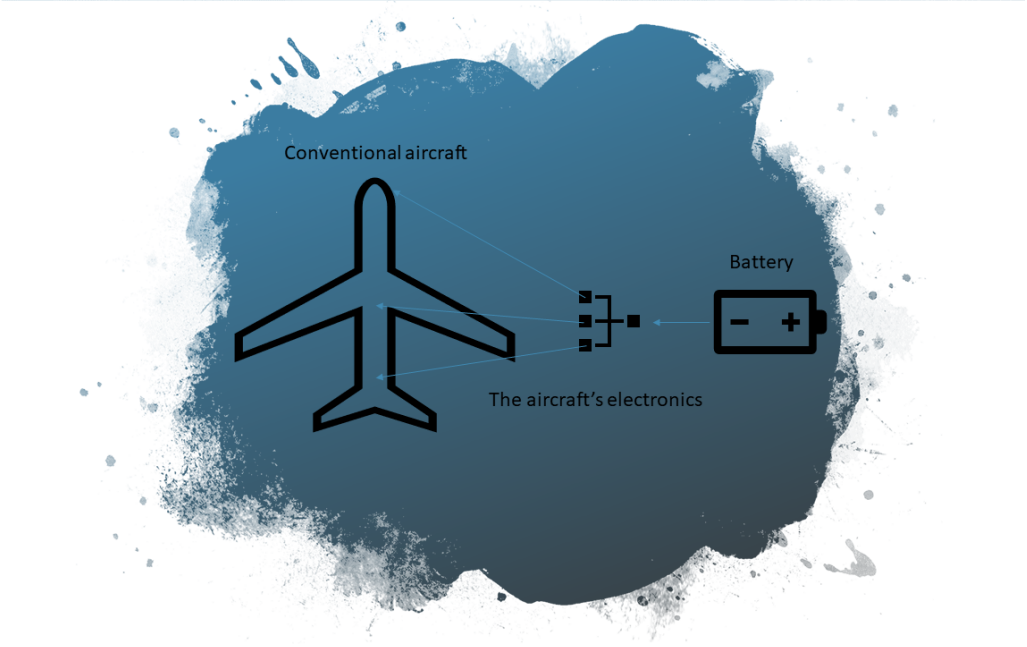


Figure 3: Hybrid electric aircraft.

FEA is currently in development for more extensive usage. However, it is already available in small market niches. Examples are urban air taxis, private transport and leisure activities (Bauen et al., 2020). In addition, according to Emmanouil (2020), the FEA configuration has been tested successfully for shorter ranges between 100-750 kms. One such aircraft is the two-seater Taurus Electro G2 by the Slovenian company Pipistrel (Xiang et al., 2018). A survey done in 2020, showed that the overall stakeholder opinion of electric aviation was positive and believed to have market value in regional flights (Mäenpää et al., 2021).

The electric aircraft receives its power from lithium-ion batteries, as can be observed in *Figure 4*. The range of a FEA configuration is significantly limited to the energy density of the battery. In the present, the batteries can have an energy density of up to 250 Wh/kg. The conventional jet fuel, however, has an energy density of around 13,000 Wh/kg, which is over 50 times as dense as that of the batteries. Furthermore, the conventional aircraft loses weight during travel, as it combusts the fuel; this enables longer flights. This is not the case for the batteries (Hou et al., 2020). Currently, there is also no standard for the charging of the batteries. In Taurus Electro G2's case, the charging process is done with the company Pipistrel's own chargers (Fuć et al., 2019).



Figure 4: Full electric aircraft.

The ideal hydrogen fuel cell consists of three components. The components of the cells are known as: the cathode, the electrolyte and the anode. Oxygen and hydrogen are being fed into the cells from one end. Water, the waste product, emerges on the other side after a steady state isobaric process, illustrated by *Figure 5*. Through this process electricity is produced (Haseli, 2018). While hydrogen fuel cells are a viable choice for powering electric aircraft, they are still in the testing stage and are far from commercialization (Bauen et al., 2020). There are a number of reasons for this. Baroutaji et al. (2019) state that there are obstacles in the form of the sustainable hydrogen having high costs, lack of hydrogen fuel infrastructure, and problems with storing the hydrogen.

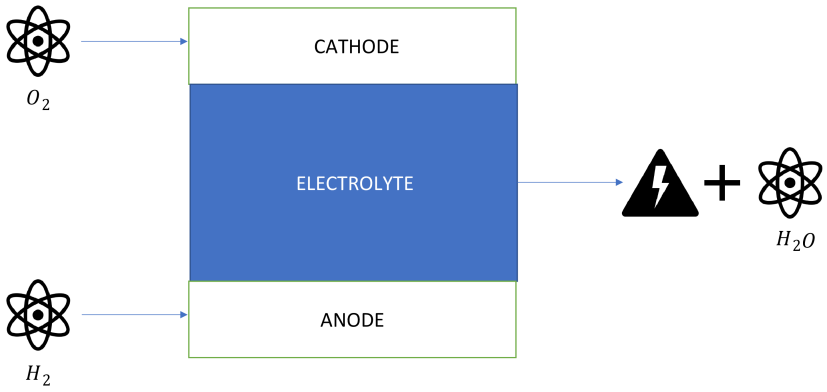


Figure 5: The generation of electricity using hydrogen.

2.2.4 Related Technological Niches

In terms of sustainability and laying the foundation for alternative fuels, LFV and the incumbent energy producer Vattenfall have recently initiated a research collaboration. The aim is to push fossil-free transportation further. This research will take place at Örnsköldsvik airport. The goal is to have the rest of the nation's airports follow the same example, should the project succeed. Here, different related innovations are being developed, e.g., automated flights and electric aircraft. The two organizations will install the charging infrastructure necessary for electric aviation. One goal is to enable and manage regional flights with ranges up to 400 kms. The electricity used in Örnsköldsvik airport will be produced by connected solar panels facilities in conjecture with batteries. This way, the operations within the Örnsköldsvik airport will become more monetarily efficient, in comparison with utilizing conventional produced electricity. The operations also become more sustainable, which reduces the climate impact. The collaboration will also include the neighboring community of Örnsköldsvik, other industries and academia to establish a holistic approach in the research (Vattenfall, 2021).

The Örnsköldsvik project is part of a larger technological niche-project called *Grön flygplats*, which is created by SRF. This is financed by the European Regional Development Fund. The project goal is to significantly reduce the emissions within the regional airports in the upcoming three years. This will partly be accomplished by fueling aircraft with biofuels instead of the conventional fuels and preparing for the charging infrastructure needed for electric aviation. Relevant actors, such as research companies and airlines, are providers of expertise and guidance to allow the goal to be

reached (Svenska regionala flygplatser, 2021).

Another niche project related to electric aviation is Elektrisk lufttransport i Sverige (ELISE). This is related to the initiative *Fossilfritt flyg 2045*. It started in 2018 at the University of Chalmers by several actors such as the KTH Royal Institute of Technology, Heart Aerospace and Uppsala University. The purpose was to bring awareness to people regarding electric aircraft and advocate for its development (Hagström, 2020). As mentioned above, the leading niche project in the Swedish development of electric aviation is Heart Aerospace and the ES-19 model (Baumeister et al., 2020). This project also collaborates with the research institutes and initiatives such as ELISE.

Lastly, networks have been created to enable a transition to electric regional aviation in Sweden. One example of such networks is The Nordic Network for Electric Aviation (NEA). It consists of major Nordic airlines, such as SAS and Icelandair, as well as the operators of airports. The actors have the aim to reduce carbon emissions in regional flight and put the Nordic countries on the forefront in the development of electric aviation (Baumeister et al., 2020).

3 Method

The methodology is divided into seven parts. The first subsection explains the philosophical standpoint and approach used in the thesis. The Second subsection describes the type of case study chosen and why it is suitable. The next subsection expands on the means of data gathering. The following subsections are the data analysis and framework analysis. These clarify how the retrieved data are analyzed and how this analysis can be used to answer the research questions. The last subsections discuss interpretivist-philosophical criteria for rigorous research as well as the necessary ethics.

3.1 Research Approach

The approach to answering the research questions was explorative. The exploration of opinions and subjective thoughts from the different actors in the field, helped identify the potential barriers and if, and in that case how, actors can affect the emergence of dominant design for electric aircraft. According to Saunders et al. (2015) a more interpretivist-philosophical standpoint is useful in handling this type of information when you receive data from verbal expressions. Moreover, the literature about the phenomenon of this emerging alternative fuel in the aviation industry is scarce, further supporting the explorative perspective (Blomkvist & Hallin, 2015). The interpretivist-philosophical standpoint is more suitable than the positivist-philosophical one. If a positivist-philosophical standpoint would be chosen, that would impede the area of study rather than enhance it. This is due to the principle of replication in the positivist-philosophical standpoint done on a specific and unstudied phenomenon such as the transition to electric aviation in Sweden (Tsang & Kwan, 1999).

Qualitative methods were chosen as the means of obtaining data. Saunders et al. (2015) also state that qualitative methods are appropriate in a research with an explorative outlook. There are multiple ways of gathering empirical data using qualitative methods. One of these methods is having a case study. As there is an ongoing transition in the aviation industry, a case study was an appropriate way of gathering data. This is because case studies allow researchers to generate insights from extensive and in-depth analysis into the study of a phenomenon, i.e., the transition of electric aviation, in its real-life context. These insights will lead to rich, analytical explanations and theory formation (Saunders et al., 2015). For further justification why a case study was used in this thesis, Yin (2003) states that a case study design should be considered when: the focus of the study is to answer "how" and "why" questions or alternatively when you cannot manipulate the behavior of those involved in the study. In this thesis, the last research question contained the interrogative word "how". Also, the writers of this thesis were independent from the decision-making processes for the development of electric aviation. This accounts for Yin's (2003) criteria for using case studies.

Lastly, an abductive approach was chosen to allow the research to be more explorative. As explained by Dubois and Gadde (2002), this is suitable when the purpose is to discover new phenomena and the aim is not simply to confirm existing theories. This also means that ideas would emerge from the gathered data, as opposed to having been generated from predetermined hypotheses. The existing literature and theories were first used to grasp the overall situation of the Swedish aviation industry. Furthermore, as it became clearer what the current situation was for the innovation and what the retrieved data were implying, the theories of literature were updated. This allowed for a deeper understanding of the potential barriers. Also, it paved the way for a proper and suitable framework in order to answer how the actors can affect the process of electric aircraft reaching a dominant

design.

3.2 The Case Study

A single case study was chosen for this research. This was also justified given the thesis aim included to comprehend the specific phenomenon of electric aviation transition within the empirical environment of the Swedish aviation industry (Yin, 2003). According to Dubois and Gadde (2014), there is still some division in academia whether to choose a single case study over multiple cases for retrieving empirical data. It was still more appropriate to have chosen a single case study here. Single case studies have historically given substantial contributions to theory and have allowed researchers to see new theoretical connections (Dyer & Wilkins, 1991). Case studies come in a variety of forms. Yin (2003) divides the case studies into three categories: explanatory, exploratory and descriptive. The explanatory case studies thoroughly examine the derived data at an extensive level. Researchers develop and test a theory based on these results. The exploratory case studies aim to investigate phenomena in the data that are of interest to researchers. They may begin with broad questions to begin a deeper investigation of the observed phenomena. Descriptive cases study natural phenomena that occur in the derived data with a predetermined theory to explain the phenomena. To fit the explorative research approach, an exploratory single case study was chosen.

The single case study was limited to the Swedish aviation industry and related stakeholders involved in the field of electric aircraft. The actors included airport managers, policy makers from various government departments, researchers from various institutes, and engineers interested in the development of electric aircraft. A major portion of the actors has directly or indirectly been involved in the development of the ES-19 model from Heart Aerospace. Heart Aerospace is the pioneer developer in Sweden with an ambition to release the innovation in the upcoming years, making it not only relevant to the thesis but also one of the only concrete examples to study in Sweden. It was possible to obtain state-of-the-art information from these actors.

3.3 Data Gathering

The use of semi-structured interviews was a key part of the methodology and the primary qualitative method for gathering the data. The aim was to ask mostly open-ended questions, where the interviewees could answer with nuance and formulate their thoughts. Some examples of questions asked can be found in *Table 1*. The motivation behind the open-ended questions was that they would enable the aviation actors to freely elaborate their thoughts. This would in turn grant extensive knowledge without compromising the core details of interest to the research questions (Blomkvist & Hallin, 2015). In this thesis, some representatives of the key actors of the Swedish electric aviation development were chosen as interviewees (see *Table 2*). These interviews with these representatives, were also the primary source of empirical data. Minor secondary data were also gathered through other sources, such as articles, journals, and webpages. The secondary data was mostly applied in the *Aviation Literature* section above. With the abductive approach, the interviews were done in three batches of 3-4 interviewees, for a total of 11 interviews.

According to Saunders et al. (2015), verifiable conclusions will be generated through the interactions between current general ideas and real-life phenomena. This was the process that this thesis undertook. From the first batch, a basic understanding was obtained of the overall situation in

Table 1: Examples of questions asked during the semi-structured interviews.

Type of question	**Asked question**
General Questions	• *What are electric aircraft in your opinion?*
Socio-technical barriers	• *What are some obstacles that you face when trying to commercialize electric aircraft?* • *Are there any existing policies or regulations that stand against the emergence of electric aircraft?* • *What business model would be appropriate for the electric aircraft?*
Financial conditions	• *What is the willingness of investors to invest in electric aircraft?*
Niche development	• *What are some ongoing projects at this time?*
Dominant design	• *How much of the development of electric aircraft is left for it to become commercially viable?*

the aviation industry – both from the perspective of the technical aspects and dynamics between different actors. Using the abductive approach, the way of thinking was updated and the questions for the interviews changed. From this second batch invaluable information was gained regarding both the pace of technological advancement, but also the in-depth information of the socio-technical dynamics in the regime and landscape. Following Saunders et al. (2015) perspective, conclusions were continuously verified with the real-life interviews. From the information gathered in these two first batches, some parts of the theoretical framework and aviation literature became inapplicable. These were subsequently updated.

When interviewing the interviewees of the third and final batch, it was quickly noticed that not much of the information was new. The matters mentioned, had already been mentioned by previous interviewees. It became apparent that the data gathering had reached a level of saturation. No further interviews were deemed necessary. The data were thus compiled and prepared for data processing. The processing itself was to be handled with a more interpretivist-philosophical standpoint where the interpretation of the content was the crucial focus (Moon & Blackman, 2014).

Table 2: Overview of interviews conducted in the aviation industry.

Batch	**Date**	**Actor**	**Position**	**Interview duration**	**Perspective**
1	25-02-2021	RISE	Researcher	40 min	Researcher
	02-03-2021	Svenska regionala flygplatser	Project Manager	35 min	Industry
	03-03-2021	Transportföretagen	Policy Expert	35 min	Policy Maker
2	09-03-2021	Heart Aerospace**	Consultant for Heart Aerospace	65 min	Researcher
	11-03-2021	Swedavia	Project Manager	45 min	Industry
	12-03-2021	ELISE	Leading Researcher	60 min	Researcher
	15-03-2021	Trafikverket	Aviation strategist	55 min	Policy Maker
2	25-03-2021	Trafikanalys	Qualified Investigator	50 min	Policy Maker
	25-03-2021	Transportstyrelsen	Environmental (Sea and Aviationfaring) Investigator	40 min	Policy Maker
	06-04-2021	NEA	Representative	60 min	Industry
	16-04-2021	Vattenfall	Project Development Manager	30 min	Industry
-	25-04-2021	KLM Royal Dutch Airlines	Sustainability Manager	*	Industry

** Received answers were obtained in a written document.*

*** Is a consultant for the company and not an employee.*

3.4 Data analysis

3.4.1 Preparing the Data

All qualitative research is based on the researchers' interactions with the interviewees. The data received can vary and be complex, dependent on how the interviewees formulate the relevant events and how these are later understood by the researcher (Saunders et al., 2015). Given this fact, it was essential that the received information from the conducted interviews was synthesized and categorized. By doing so, it became easier to draw conclusions for the research questions.

After each interview in the case study, the notes were directly reviewed, while the key elements of the conversation were still fresh in mind. Next, the audio-recorded conversations were transcribed by a transcription program. In this case, my.sonic.ai and Nvivo were both used. The machine transcribed transcripts were then manually corrected for various errors. This process was done to avoid bias by simultaneously listening to the audio file. The corrected and finished transcripts were analyzed through the program Nvivo. The coding was done manually to avoid the potential fragmentation that comes with automatic coding (Auld et al., 2007). By coding the transcripts which can be observed in *Table 3*, the generated ideas and concepts that were previously unorganized, complex and substantial, were transformed into data segments (also known as codes or nodes in Nvivo). In this way, the information was much more manageable (Bazeley & Jackson, 2013). Important to note is that the data have not been forced into a particular number of codes (Saunders et al., 2015). Instead, all possible meanings of the data were explored. This determined the number of codes, which was 21. This particular part of the data analysis was done with a more inductive approach.

Table 3: Codes/nodes used in Nvivo.

Specific role of electric aviation	General measures	Motivation for implementing electric aviation
The different roles of all alternative fuels	Business models	Budget
Political	Regulations	Cooperation
Other socio-technical aspects	Technical aspects	Timeline for alternative fuels
The appearance of electric aviation	What electric aviation is	The public opinion
Certification	The present state	The supplier perspective
New actors	Reactions from incumbents	Other important information

3.4.2 Analysis

The more inductive part of the analysis was done using a thematic analysis. This is a method for analyzing qualitative data in a systematic way. Thematic analysis is only an analytical tool and not

a methodology. It is flexible given it is not based on a specific philosophy. It can be used to analyze large and smaller amounts of data (Braun & Clarke, 2006). It can also help highlight meaningful elements of the data (Clarke & Braun, 2016). According to Saunders et al. (2015), the search for themes only begins when the coding process is complete. The thematic analysis should be used to create a short list of themes related to the research questions. Searching for these themes involves making judgements about the data and how everything fits together for further analysis. Next, it is important to test the propositions regarding the themes that appear inductively. After rigorous testing, valid and well-grounded conclusions can be drawn.

In this case, thematic analysis was used to analyze the codes/nodes from Nvivo. Patterns were identified and themes gradually appeared. Initial themes were questioned and some themes were combined. Important elements were for instance the possible barriers. After adding the relevant content of codes/nodes into the themes, these were then questioned. An example would be the actor collaboration. As opposed to the perception of the interviewees, alternative explanations were sought. In the end, a different conclusion regarding this theme was found. This process was continued while iteratively condensing the themes. These themes can be found in *Table 4*. In the end, the thematic analysis allowed initial answers for the first research question to be drawn.

Table 4: Finalized themes of the thematic analysis.

Theme	**Description**
Financial conditions	Perceived financial barriers and proposed measures.
Regulatory & Political conditions	Perceived regulatory or political barriers and proposed measures.
Business Model & Legitimacy Conditions	Perceived barriers in business activities and legitimacy of electric aircraft and proposed measures.
Technical Conditions	Perceived technical and certification barriers and proposed measures.
Actor Collaboration	The communication, cooperation and collaboration between the actors.
Timeline	Information about the set dates for the release of alternative fuels in Sweden.

The second part of the analysis was to connect the empirical data to the conceptual theories. Conclusions could then be drawn regarding the second research question. An explorative framework was the means of directing the analysis of the empirical data. This is also according to how Saunders et al. (2015) suggested to perform a more deductive analysis. Here, systematic combining was chosen as an appropriate method to isolate the relevant theories in the framework. This process was nonlinear. The focus constantly moved between the framework, the data, and the analysis, as explained by Dubois and Gadde (2014).

The conceptual theories used in the analysis were mainly MLP and dominant design theory. The main source of inspiration was Geels (2006), Geels (2012) and Yap and Truffer (2019). The MLP was used to map out all important components of the transition into the three analytical levels of *niche innovation*, *socio-technical regime*, and *socio-technical landscape*. These components included among other things important niche innovation projects, the various actors and the identified barriers. Using this mapping of the transition, patterns were determined. To explain the pattern observed in the data, a new concept was introduced – the *"socio-technical gate"*. The concept explains the connection some of the barriers had in common. Passing through this conceptualized "gate" would make the related barriers easier to overcome. Lastly, dominant design theory was used to explain how the findings could explain how the actors can affect the emergence of the dominant design. A clarification of all the theoretical concepts can be found in *Table 5*.

Table 5: Theoretical concepts and examples of their operationalization in the thesis.

Theoretical concept	**Description**	**Examples of its operationalization**
Niche innovation	The analytical level where niche innovation projects develop. This also includes niche markets and technological niches (Geels, 2006).	Projects such as *Grön flygplats* and *ES-19*.
Socio-technical regime	The analytical level where cognitive routines and activities of social groups that maintain and change the elements and linkages in socio-technical systems (Geels, 2006).	Various actors such as airports, airlines, and researchers.
Socio-technical landscape	The analytical level that covers the wider context of a socio-technical system. Consisting of cultural or normative values as well as material arrangement. Occurring changes in this analytical level can create pressure on the socio-technical regime, allowing for window of opportunity (Geels, 2006).	The cultural values include *flight-shame*. Political pressure refers to policies such as the zero-net greenhouse gas emission goal for 2045. The physical infrastructure includes the electricity grid.
Barriers	The obstacles or hinders in the way of development. In this thesis this refers to the barriers for the development of electric aircraft.	Four categories of barriers were identified. These being the *financial*, the *business model and legitimacy*, the *regulatory and political*, and the *technical and certification*.
Dominant Design	A single product design that achieves market dominance (Srinivasan et al., 2006). This design is used as the template for all products in the same category.	The design of electric aircraft that all subsequent products would be templated after. This design has not been set yet.
Dominant Category	A theoretical concept or a socio-cognitive construct between stakeholders (Bingham & Kahl, 2013).	In this case, it symbolizes "what" electric aircraft is. Current categories include "battery-powered", "hydrogen fuel cells", "having an electric engine" or "hybrid aircraft". These will converge into a single dominant category.
Window of Opportunity	Theoretical period of time which starts with the convergence of the dominant category and ends with the emergence of the dominant design (Suarez et al., 2015).	The period of time of which commences at the convergence of the dominant category (which in this case is "electric aviation") and ends with the emergence of the dominant design of the electric aircraft.
Socio-technical Gate	A new concept introduced in this thesis that encapsulates the connection between certain barriers. By metaphorically passing through this gate, related barriers would become easier to be overcome.	The first *socio-technical gate* in this thesis is actor collaboration and the second *socio-technical gate* is the legitimacy of electric aviation.

3.5 Research Quality

The quality of case studies with a more interpretivist-philosophical standpoint cannot be easily and properly assessed by examining and addressing positivist-philosophical research quality criteria, e.g., presented by Gibbert et al. (2008). The criteria being put forward are *Internal validity*, *Construct validity*, *External validity*, and *Reliability*. Traditional criteria of validity and reliability are not applicable in the same way for a more interpretivist based qualitative research (Shah & Corley, 2006). Instead, the *Trustworthiness Criteria* from Lincoln and Guba (1985) were used in the thesis. These criteria are the following: *Credibility*, *Transferability*, *Dependability*, and *Confirmability*.

Credibility stands for the level of confidence in the truthfulness of the findings. This is usually addressed through having conducted thorough literature reviews and checking of received information (Connelly, 2016). In this case, *Credibility* was addressed through the means of having an extensive engagement in the field of the subject in question and utilizing triangulation. A thorough review of the literature on the topic of electric aviation was conducted, as well as active communication with various stakeholders about the innovation, in order to stay well informed and engaged. To achieve triangulation, multiple perspectives were considered, where the derived findings from one interviewee would be ensured for correctness by cross-checking them with the data from the other interviewees (Shah & Corley, 2006).

Transferability is to what extent the findings are beneficial and transferable for researchers in a different setting. Here, it is the readers of the thesis that determine how useful the findings or processes are to their own work (Connelly, 2016). Therefore, it is difficult to properly address this criterion. Nonetheless, it is possible to achieve the criterion of *Transferability* by having transparent reporting as stated by Kyngäs et al. (2019). In this case, this was done in discussion with peer groups. The processes or methods used in the research and information received from interviews were presented and therefore assessed by the peer group whether the techniques used in this thesis could be replicable. Furthermore, all methods and results of relevance to the thesis have been clearly stated in order to maximize the *Transferability* to other settings with similar conditions, such as another country's aviation industry.

Dependability alludes to the research's stability of the conditions it was conducted in, i.e., whether the study can be replicated with the same participants under the same conditions (Cope, 2014). There are mainly two procedures for addressing *Dependability*. This first is to have a well-documented and consistent process, e.g., describing the interviewees and writing a coherent thesis. The second is to have peer-briefing with other colleagues (Connelly, 2016). Following the first procedure, the aim was to have clear descriptions, especially in the methodology section, through the entire research process. After writing each new section, its relevance to the rest of the thesis was assessed. In the case that prior sections would become irrelevant with the new writings, these would be modified to have the thesis stay coherent. In addition, all actors that were interviewed during the thesis have been carefully documented and presented. This approach is cohesive with Kyngäs et al. (2019) suggestion of how to address this particular criterion.

Confirmability refers to the objectivity of the research findings and that it is not tainted by bias. A method to address this criterion is by documenting how conclusions and data interpretations were determined (Cope, 2014). This has been done through three means similar to what is presented by Shah and Corley (2006). Firstly, the data retrieved from the interviews have been diligently

recorded in high quality, both in audio and video format. In addition, the recorded files have been carefully managed and categorized after each interview. Secondly, the transcripts have been corrected manually after each interview with either the audio or video source. Thirdly, an explicit distinction has been made between first and second order findings. The first order findings being the corrected transcripts translated into codes in Nvivo while the second order was the thematically analyzed codes.

3.6 Research Ethics

When conducting a thesis, it is important to adhere to the following four guidelines to ensure a good research ethic: *informing about the thesis purpose*, *having the consent of participants*, *being confidential with the information received*, and *solely utilizing the information for research purposes* (Vetenskapsrådet, 2002). The first guideline was addressed through having a minor briefing before the interviews regarding the purpose of the thesis and the interview. The second guideline was addressed when the interviewees accepted to participate. Regarding confidentiality, none of the participants' names or personal details have been presented in the thesis. Lastly, the gathered data have been used for research purposes only.

Furthermore, it is critical that the researchers display competence, transparency, and honesty when approaching an actor. It presents the researchers as impartial, which allows for easier data gathering (Saunders et al., 2015). This was carefully executed with the actors of the aviation industry. Prior to the interviews, up to date literature was carefully read and reviewed. During the interviews, authentic expertise in the subject of electric aviation and chosen theories were communicated. Whenever a topic could not be clarified, e.g., while asking a question, the knowledge limitation was directly and transparently stated to show honesty. In addition, it is important to prevent any misrepresentation and eliminate inaccurate reporting when acquiring and gathering data from the qualitative case study (Bell & Bryman, 2007). The answers of the interview questions were thus always discussed and reaffirmed with the interviewee during the interviews.

4 Results

The results section is divided into six parts. The first four subsections present the actors' answers to semi-structured questions. It contains their perceived barriers as well as proposed measures for the identified socio-technical conditions from the thematic analysis. The subsection 4.5 summarizes these findings. Additional relevant findings are presented in subsection 4.6. This contains the actors' thoughts of the collaboration between all actors as well as the temporal aspect. All the subsections contain tables consisting of data from the interviews.

4.1 The Financial Conditions

4.1.1 Financial Barriers

The view on how to allocate resources for the development of electric aviation is varied, which can be observed in *Table 6*. From the industrial perspective, the implementation of new infrastructure is a huge undertaking. Both regional and national airports need more financial support. The regional airports are currently relying on operational fees. It is also unclear whether the municipalities owning the airports are willing to bear the costs, since they also have other projects to consider. Investments from outside sources would be welcomed, since the whole industry is in a crisis. The state-owned airports on the other hand, have negotiated with aviation companies to jointly invest in the structural changes. The funding depends on the attitude these companies have toward the development of electric aviation. There are seemingly no problems with the funding for the state-owned airports. The state-airports would, however, gladly accept other investments to decrease the burden on the aviation companies.

All the researchers, policy makers and the general industrial perspectives are similar in terms of the overall financial conditions. All of them also see the funding as a decisive limiting factor. According to some of the researchers, the development can be improved with more resources. They see a clear lack of economic support, from both the industry itself and society as a whole. Investors have adopted a wait-and-see strategy, without providing enough funding for the research. This has also slowed down the research considerably.

Table 6: The financial barriers.

Industry	• *"Significant investments are necessary to get this charging infrastructure." – SRF, March 2021* • *"There are reasons to get help from the state to develop the infrastructure and get started. New investments are very high and not everyone has the opportunity to put the sum directly." – Swedavia, March 2021* • *"We imagine that it would be quite expensive, because R&D for developing the first models is very costly." – KLM Royal Dutch Airlines, April 2021*
Researchers	• *"What is clearly the biggest obstacle is that you do not allocate sufficient resources in this phase. And that is from both society and industry." – ELISE, March 2021* • *"One thing is definitely the financing of it. Because we are a number of years away from when you will be able to start making money from this." – Heart Aerospace consultant, March 2021*
Policy Makers	• *"The finances is always a factor, an obstacle, no matter what innovation or research you are talking about." – Trafikverket, March 2021* • *"The biggest obstacle now is the finances. That the entire aviation industry is bleeding, that is obviously the biggest problem." – Trafikanalys, March 2021*

4.1.2 Proposed Measures for the Financial Barriers

As shown in *Table 7*, the financial barriers can potentially be overcome by the authorities through loan guarantees, implementing discounts for the electric aircraft or research funding. This is especially the case for the initial phase of the introduction of electric aircraft. Similar to other projects, there is a need for initial support. One interviewee explained that there must also be a buyer for the aircraft. As these can have both substantial initial and maintenance costs, some form of financial support is not only needed during the development, but also after the aircraft reach the market. Lastly, one researcher rationalized that if the research is valued by the authorities, then they should try to fund it. If an analysis would show that it is possible to accelerate the research and reduce the research duration to just one year with a billion SEK, then it should be worth the cost.

Table 7: The proposed financial measures.

Industry	• *"We have applied for and received grants from the European Regional Development Fund, which co-finances [...]" – SRF, March 2021* • *"At Swedavia's airports, we have invested in electric flights. It is an internal negotiation between Swedavia and the flight operators. Of course, if there is a grant or support for the green transition, then it is obviously something we are interested in." – Swedavia, March 2021* • *"Airlines could and should invest in new aircraft themselves." – KLM Royal Dutch Airlines, April 2021*
Researchers	• *"You need to work a lot with increased support for the development of this type of technology." – RISE, February 2021* • *"You can compare it with anything, if you were to start investing billions in this today, then you would make this work pretty soon. It itself, cannot be considered a major cost, if for example, you compare it with building a new construction project or build train rails." – ELISE, March 2021*
Policy Makers	• *"Then it is a matter of supporting this development, for example through and from the research funding authority. This is the foundation." – Trafikanalys, March 2021* • *"Then when it comes to the market; someone has to buy the plane. [...] Here you need to give some form of support for the introduction of these planes. [...] It could be loan guarantees." – Trafikanalys, March 2021* • *"Here we think that incentives are needed for the first aircraft that are powered by electricity, such as discounts." – Transportföretagen, March 2021* • *"The state and other financiers must see to it, that money is put into that pot." – Trafikverket, March 2021*

4.2 The Regulatory & Political Conditions

4.2.1 Regulatory & Political Barriers

There are no direct laws or regulations that would hinder the development of electric aircraft. The concept is far too new for this to happen. Instead, existing regulations take aim at conventional fossil-fueled airplanes. The developers and researchers believe that these regulations need to be optimized to better suit future electric aviation. The industry itself, specifically the airports, is of the opinion that the industry actors will not be too affected by any regulatory changes. They are simply following the current regulations regarding passengers. From the policy makers' point of view, the most important thing with the regulations is risk-control. It is necessary to determine the risks of having electric aircraft, such as high-voltage and potential battery-fires. The different types of actors have the same views on politics. They all desire more support from the government in the shape of establishing clear goals, laws, economic support and tax reductions on the purchase of electric aircraft. This is also illustrated by *Table 8*.

Table 8: The regulatory and political barriers.

Regulations	
Industry	• *"We try to investigate the security rules that are needed for the electric aircraft." – Swedavia, March 2021* • *"The currently applied rules for passengers [...] – both nationally and internationally, will not change. The regulations will not affect us." – Swedavia, March 2021*
Researchers	• *"The current regulations for airspace could be more optimal for electric aviation, this needs to change." – ELISE, March 2021* • *"Things in regulations and similar, need to be updated to facilitate the introduction of new technology." – RISE, February 2021* • *"In the present, there are no regulations for how to handle electric aviation." – Heart Aerospace consultant, March 2021*
Policy Makers	• *"It is no longer about how to handle reserve charging stations with fossil fuels, but how to handle electricity. [...] Also, how to deal with the new risks that comes with the high voltage and battery fires." – Trafikanalys, March 2021* • *"I think that policies are going to be necessary to drive the development forward and get a commercial product." – Transportföretagen, March 2021*
Politics	
Industry	• *"Everything comes down to what the industry gives and the flight operators find interesting. If they do not find this interesting, then incentives from the government may be necessary." – Swedavia, March 2021* • *"I am afraid that everything will end up being just talk and nothing else. We've already talked for 6-7 years. We no longer have the time to continue talking." – NEA, April 2021*
Researchers	• *"Then you will, from the politicians, need some regulations or support to continue this." – RISE, February 2021* • *"A new industrial project that could potentially generate earnings from export to Sweden. [...] A new market in the whole world with an exciting new airplane. They show pride that there is an electric aviation project in Sweden. But they do not invest any capital in it." – Heart Aerospace consultant, March 2021*
Policy Makers	• *"We can drive the development in other ways, where the EU changes the regulations so that you do not have to be technology neutral, but instead pro-electric aviation." – Trafikverket, March 2021* • *"Sweden needs aviation, due to its geography. It is important to have good opportunities with aviation and changing the current aviation industry. [...] making it a bigger political question, [...] like in Norway." – Trafikverket, March 2021*

4.2.2 Proposed Measures for the Regulatory & Political Barriers

It is of vital importance that the rules for electric aviation are set by decision makers, but it is even more important to first set a standard for the charging infrastructure, as seen in *Table 9*. From there,

it will be easier to set the regulations in terms of safety measures and security rules. Focus should be put on the overall development of the aircraft, certification and charging infrastructure. Political support is necessary, according to all types of actors. The policy-makers suggested a number of political instruments and measures to facilitate the introduction of the aircraft, e.g., the authorities should purchase the first-generation electric aircraft and then rent these out. Ideally, such measures will support smaller start-ups financially.

Table 9: Proposed regulatory and political measures.

Regulations	
Industry	• *"Then it is not about rules, but rather that a standard is needed. To know how it is supposed to be charged and that multiple charging infrastructures are not needed for different aircraft." – SRF, March 2021* • *"We might not even get to fly these aircraft commercially, due to not having completed the standardization or certification. That is why both of these aspects need to be developed at the same time." – NEA, April 2021*
Researchers	• *"But there are many other things that need to be fixed, such as certification and rule works to get permission to fly the aircraft in the air and on the ground." – ELISE, March 2021* • *"It will be the aircraft developers and their partners that will have to develop the charging infrastructure. Neither the state nor the airports will do so." – ELISE, March 2021*
Policy Makers	• *"A solid example within policies that can be done today is to remove the aviation taxation for electric aviation or to reduce the landing fee on the airports. That is a green incitament that changes the margin" – Transportföretagen, March 2021*
Politics	
Industry	• *"It is partly incentives. But, to develop this when it is new, there might have to be a political will." – SRF, March 2021* • *"The aviation industry has always had high peaks and low valleys. There is a very low valley right now, where it is hard for the flight operators to motivate doing this – when there are uncertainties around the technology and business opportunities. Continuous help from decision makers will be necessary." – Swedavia, March 2021*
Policy Makers	• *"A number of different policy instruments, a number of measures to facilitate the introduction. It is partly for long term goals. The governmental powers need to set goals for what is necessary. It creates a sense of safety for those that drive the development forward." – Trafikanalys, March 2021* • *"The bigger companies are not interested in buying first-generation electric aircraft. [...] It is therefore often the smaller companies that are interested and they do not have the financial capabilities to buy new electric aircraft. It is a very big investment for them. [...] here the state could buy them up and rent them out." – Trafikanalys, March 2021*

4.3 Business Model & Legitimacy Conditions

4.3.1 Business Model & Legitimacy Barriers

Policy makers are of the opinion that the large incumbent aviation companies have no desire to incorporate the first-generation electric aircraft into their businesses. This can be observed in *Table 10*. It comes down to three major factors. Most incumbent aviation companies already have an aviation fleet, in which the smaller electric aircraft will not fit. According to researchers, the second factor is that their business models are adapted for longer ranged flights, as it is more efficient from

an economic standpoint. This makes them unsuited for the first-generation electric aircraft that are only capable of making short-range flights. Thirdly, companies distrust the completely new technology, e.g., the lithium-ion batteries. In addition, according to one interviewee, it is much easier to transition from fossil fuels to biofuels, given that it does not require anything other than changing the fuel in the jet-engine. Electric aviation on the other hand needs substantial change in airport infrastructure and investing in whole new planes.

In the case of legitimacy, which includes the public opinion of electric aircraft – a majority of the interviewees stated that it is difficult to know what the public thinks. The technology has recently emerged and has not been tested commercially. A common opinion shared among the industry actors, and more specifically the airport actors, is that new technology needs to be embraced. One must believe in it, even if it is new. According to them, there are people that do not see any future in battery-powered electric aircraft technology. Nonetheless, some believe that as soon as electric aircraft start to operate commercially, a positive opinion of the aircraft will emerge. Another issue is the phenomenon known as *flight-shame*. One researcher explained that it was due to the climate impact of conventional aircraft emissions. Should sustainable alternatives emerge, then this might change. Also, several actors stated that safety is paramount to create a positive public opinion.

Table 10: Business model and legitimacy barriers.

Business Model	
Industry	• *"Small companies have smaller airplanes, with a fitting fleet. Last week however, Finnair agreed to buy such smaller airplanes from Heart Aerospace. But of course, it does not really fit into SAS's fleet and their business model." – NEA, April 2021* • *"An obstacle that I would say is – it is easier to have a transition with biofuel into the jet engine – since with electric aviation you have to change the infrastructure in part, but also buy a new aircraft." – Swedavia, March 2021*
Researchers	• *"It is cheaper with larger fossil fueled aircraft, both because you share pilot costs and because the internal combustion engine has a higher efficiency if it is larger. That is why the trend has leaned towards larger aircraft. " – ELISE, March 2021* • *"You may need to look at new business models and other aspects that may need to be rethought, to realize this commercial electric air traffic in the future." RISE, February 2021*
Policy Makers	• *"My opinion is, of course, that electric planes must fit with the traffic they are to operate in. You are right that SAS has a fleet of larger aircraft." – Transportföretagen, March 2021* • *"The larger companies are not interested. SAS is not interested in flying first generation electric aircraft. Because they do not fly at these small airports with few passengers, but operate larger routes and have a fleet adapted for this. It is often smaller companies that are relevant and they do not have large funds that can buy new electric aircraft." – Trafikanalys, March 2021*
Legitimacy	
Industry	• *"There are voices pointing out that there is no future with battery-powered electric aircraft. That the batteries are too heavy and have too little effect and that one should invest more in hydrogen and biofuel. This may be true to some extent, but one should not be categorically dismissing electric flights altogether." – Swedavia, March 2021*
Researchers	• *"Yes, it is a very new technology. There is nothing for the public to see. [...] Then you have the fact that things like flight-shame, which has popped up over the past year. Especially here in Sweden where they actively choose to not fly." – RISE, February 2021* • *"Then, you have actors who say that flying itself by definition is bad, that flying is an incredible waste of energy compared to anything else – the train is so fantastic." – ELISE, March 2021*
Policy Makers	• *"It is difficult for us to say how it will be perceived" – Transportföretagen, March 2021* • *"There is this need for acceptance, that it is safe to fly with electric aircraft." – Trafikverket, March 2021* • *"We do not see the need to transport oneself regionally." – Trafikverket, March 2021*

4.3.2 Proposed Measures for the Business Model & Legitimacy Barriers

The situation of electric aviation is uncertain, as can be observed in *Table 11.*, but it is also an opportunity for smaller players to establish themselves on the market. The larger airlines will have difficulties in adopting the first-generation commercial aircraft. They will not be very interested in these, due to having different established business models for their fleets of bigger aircraft – it is infeasible for them to suddenly switch to smaller aircraft. According to one researcher, having electric aircraft landing and taking off at smaller airports can form a more distributed node system with more frequent flights and simpler airplane transits. There is a great potential for new suitable business models using this node system. What is left, is to reassure the public by having the authorities review everything before commercialization. The feedback is expected to have a positive outlook as long as there is a sample of a successful product. The actors therefore state that one aviation actor should go ahead and introduce electric aviation to the market, thereby showing that it is feasible. This would strengthen the legitimacy of electric aviation.

Table 11: Proposed measures for business model and legitimacy.

Business Model	
Industry	• *"But it is also an opportunity for smaller players to establish themselves in a niche in the market, that the larger players do not have the ability to adapt to today." – Swedavia, March 2021* • *"The existing airlines may not be those who are interested in this first electric aircraft and then it will have to be some new players, who dare to take on the new aircraft." – SRF, March 2021* • *"Of course, transitioning to a more sustainable aviation sector, does also mean looking into new business models. Electric aviation [...] or other alternatives could be part of this solution." – KLM Royal Dutch Airlines, April 2021*
Researchers	• *"So maybe these new types of actors are needed since they can fully invest in making this happen and work." – RISE, February 2021* • *"If you can get the right financial conditions using small aircraft taking off and landing at smaller airports. Then, you can suddenly think of it as a more distributed node system. [...] With more frequent departures and easier transits, then suddenly you can fly on nodes instead [...]. Maybe you would only have to stop 1-2 times when traveling to Southern Europe." – ELISE, March 2021*
Policy Makers	• *"In the long run, electric aviation will be established, due to it being a viable solution for shorter flight distances. There is no potential in the aircraft currently being developed [...] unless the batteries are developed a bit more." – Trafikverket, March 2021* • *"There is no commercial production of aircraft under 50 seats. That is the size suitable for travel to Gotland, Ronneby, Kalmar, inner Norrland." – Transportföretagen, March 2021*
Legitimacy	
Industry	• *"We try to explain what the airports do, if we get the opportunity [...]" – SRF, March 2021* • *"One must dare to believe in technology – when technology is new." – SRF, March 2021*
Researchers	• *"It requires someone to go in and really invest and show that it works. That people have tested it now and that it feels completely normal to use it." – RISE, February 2021* • *"If you can get a commercial line where people fly and perceive it as much smoother and more comfortable with electric flights, then I think you can start to change the whole thing." – ELISE, March 2021* • *You need a blowtorch, something that goes before and shows this works in practice. There are such examples historically." – ELISE, March 2021*
Policy Makers	• *"What I think can reassure the public is that everything you put on the market is reviewed by the authorities." – Transportföretagen, March 2021* • *"If it has been shown that it works, then the first step to general acceptance has been reached." – Trafikanalys, March 2021*

4.4 The Technical Conditions

4.4.1 Technical & Certification Barriers

As can be observed in *Table 12*, a variety of different answers were given regarding the technical barriers of electric aircraft. The representatives of the industry were concerned with the electricity supply. It was unknown whether the Swedish electricity production could handle the eventual rise of electricity demand, due to the introduction of electric aircraft. Especially, if multiple aircraft needed to be charged at once, creating a peak demand. A worry was regarding the morning and night peak travel points. These peaks of demand would perhaps strain the power grid.

Another identified technical barrier is the current level of energy density in the batteries. The batteries are sufficient for shorter range travels. More development will, however, be necessary to enable major commercial usage. Though the development of the batteries, aerodynamics and charging infrastructure is important, the major work lies in the integration of the different parts. It is not a matter of developing more advanced technologies, but rather to combine existing parts and technologies into a new electric aircraft. Lastly, the certification process is not a simple matter. No standards exist for the electric aircraft to follow for certification, since the aircraft is not finished yet and there is no working example on the market.

Table 12: Technical and certification barriers.

Technical	
Industry	• *"The airports are trying to prepare for when a standard is ready" – SRF, March 2021* • *"The big challenge is to have [...] the infrastructure and all in place." – NEA, April 2021* • *"Among other things, there must be a backup energy reserve on airports. Which almost always is diesel powered." – SRF, March 2021* • *"It is easy to fuel a conventional plane, where the only time-consuming factor is the passengers. With electricity, the charging infrastructure and power grid needs to face the demand. [...] Which requires fast charging." – Swedavia, March 2021*
Researchers	• *"The amount of energy is not a problem, however if all of them need to be charged at once, then it is not a trivial matter to solve." – ELISE, March 2021* • *"Charging infrastructure, battery density, aerodynamics. All of those things are important, but the majority of the work is about integration. To use existing technology and piece it together." – ELISE, March 2021*
Policy Makers	• *"It is not rocket science, the scaling-up is the problem." – Transportföretagen, March 2021*
Certification	
Industry	• *"The certification process takes a long time, due to the fact that the regulations do not exist for this kind of aviation, since the aircraft have not been developed yet." – SRF, March 2021*
Researchers	• *"Other issues are certification and regulations" – ELISE, March 2021* • *"One of the greater challenges is to build something that is also certifiable. Which is why Heart Aerospace have taken the approach to work with it from the start." – ELISE, March 2021*
Policy Makers	• *"Pipistrel has an EASA certified electric aircraft. The certification process was quick back then. The process now is the same, but there are more components to check and the aircraft is bigger." – Trafikverket, March 2021* • *"It should go through the same measurements as ordinary airplanes and fulfill all conditions. Only after this, you have a product. That is the legal demand. There is an issue, however, that there is no standard. EASA has certified smaller, two-seated airplanes, but it is different from certifying a larger electric aircraft." – Transportföretagen, March 2021*

4.4.2 Proposed Measures for the Technical & Certification Barriers

In terms of measures for technical and certification barriers the opinions are scattered, as illustrated in *Table 13*. The policy makers perspective is not included given the topic is not within their expertise. One interviewee believes that there will be a jump in the field of batteries technology, making the batteries much more efficient. According to the researchers, there should not be a problem to reach distances up to 400 kms, using the present technology of electric aircraft. The

engineering work can be completed with the help of projects such as ELISE. By using the optimal trajectories for flight, up to a 100 kms can be gained. When it comes to the technical problems with the power grid, this will not be an issue in the short term. Depending on the expansion, there might appear some bottlenecks however. This is not due to insufficient power, but rather the distribution.

Several actors have pointed out that the certification process is not trivial and might prove to be a barrier. To certify a completed electric aircraft from scratch could prove to take time, which might be detrimental for the introduction. The certification process, however, started at the same time as the development of the aircraft started. This means that these two processes are done in parallel. By certifying it in parts, rather than a whole aircraft, makes the whole process faster.

Table 13: Proposed technical and certification measures.

Technical	
Industry	• *"Many are researching batteries. I believe that there will be a 'jump' in technology that will make them much more efficient than they are today [...]. When this aircraft starts flying, maybe 2026-2028, then there might be other batteries available than those being tested now." – SRF, March 2021* • *"With how electric aviation looks today, their usage being in school flights and transportation of goods, there are no issues with the power grid. Of course, there might be times when it is tight with power. At those times, we might have to use our electricity storage to a certain degree to combat the problems." – Vattenfall, April 2021* • *"When electric aviation becomes more regular, we might see some bottlenecks in certain locations. The power grid infrastructure and local production capabilities will need to be investigated." – Vattenfall, April 2021*
Researchers	• *"We are looking at electric engines with high energy efficiency, high effect, and low weight. With my estimations, 400 km range is fully doable with today's battery technology. Some engineering is necessary to make it work. That is also where ELISE helps accelerate the development of regional flight in Sweden." – ELISE, March 2021* • *"From an energy consumption perspective, you would want to fly according to the best trajectories, where you glide much to the end to save some percentage of fuel. For electric aviation that approach is important, as up to 100 kms can be earned with the right way of flight. Which is significant when discussing 400 km ranges." – ELISE, March 2021*
Certification	
Industry	• *"They started the certification process when they had just begun developing the aircraft. Just to make the aircraft development proceed in parallel with the European certification." – SRF, March 2021*
Researchers	• *"You do not have to certify a whole complete aircraft, doing it in pieces should be alright. First the driving system, then certification of the cabling or the charging, then it is the wing construction and so on. Smaller project parts like this, until we have a complete aircraft. It is another way of working." – Heart Aerospace consultant, March 2021*

4.5 Summary of Perceived Barriers and Proposed Measures

4.5.1 Summary of the Perceived Barriers

The perceived barriers for the development of electric aviation were categorized into four of the themes identified in the thematic analysis. These were then the *Financial*, *Regulatory & Political*, *Business Model & Legitimacy*, and *Technical*, as illustrated by *Figure 6*. Within the theme of *Technical* barriers, there are the inadequacies of battery technology, problems with certification and assembling the electric aircraft from different subcomponents. The *Financial* barriers are divided into the lack of funding for the development of electric aircraft and connected infrastructure to these aircraft. In *Regulatory & Political* barriers, the issues included the non-existing regulations for potential risks in the electric aircraft, i.e., safety measures or security rules. This also included the

lack of political support and clear goals by the authorities. Lastly, the barriers of *Business Model & Legitimacy* theme can be divided into the electric aircraft being unsuitable for the incumbents' business activities and the lack of faith in the innovation.

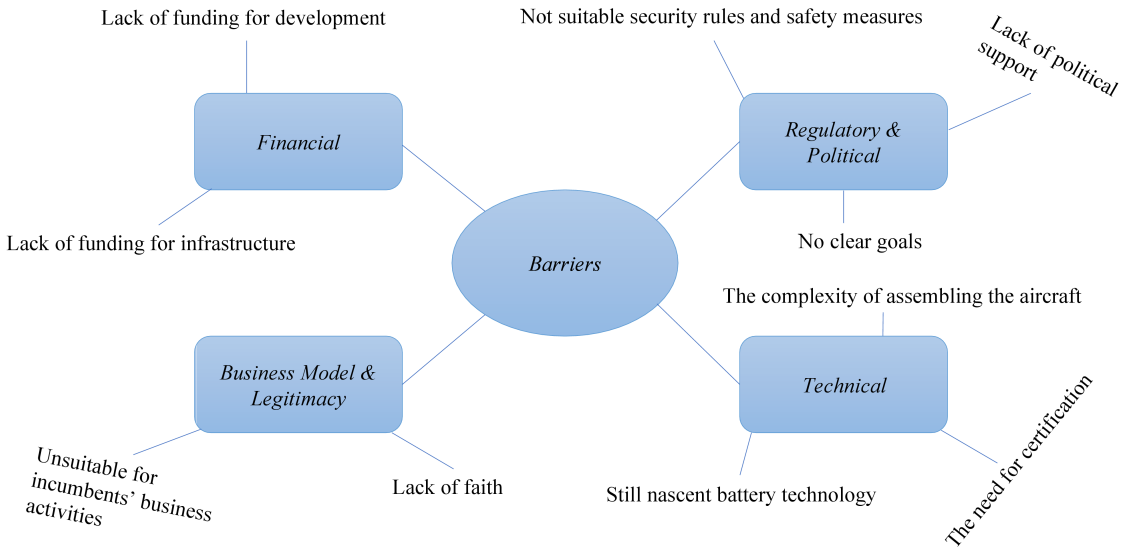


Figure 6: Mapping the actors' perceived barriers.

4.5.2 Summary of the Proposed Measures

For the financial barriers, proposed measures include three courses of actions. One measure was to let airlines bear their own funding. On the airport side, funding can be secured by grants from sources such as the European Regional Development Fund. On the state-owned airports, funding can also, to a certain degree, be secured from aviation operators. The last course of action was directed mostly at research and newcomers. It was to let authorities support the introduction of electric aircraft, through means such as loan guarantees, funding for research, and discounts for electric aircraft.

For the political and regulatory barriers, measures include creating regulations for risk-control. This includes the various risks of the electric aircraft and the necessary infrastructure, such as high-voltage and dangerous battery-fires. Most actors also desire more support from authorities in the shape of clear goals, laws, financial support and tax reductions on purchase of electric aircraft.

For the business model barriers, where the new aircraft may not fit the established business activities, aviation companies are encouraged to formulate new business models. A measure here was to form a more distributed node system, where more frequent flights and simpler airplane transits would be possible. The legitimacy barrier affects both the public and potential investors. Having the

authorities review the technology before the commercialization was suggested. Multiple actors suggested a sample of a successful product. Proving its feasibility was expected to lead to positive feedback from the market.

No actual measures were suggested for the technological conditions. A jump in the technology was, however, expected from the battery side. The representative of Vattenfall stated that there should be no issues with the power generation. A measure already taken for the certification process, was to certify in parts, rather than certifying whole aircraft.

4.6 Additional Findings of Significance

4.6.1 Actor Collaboration

Generally, the actors believe that the collaboration and communication between the different actors for the development of electric aircraft is quite good. This can also be observed in *Table 14*. Most actors do show clear willingness for collaboration and some collaboration has already started. Airports collaborate with electricity producers and aircraft developers collaborate with companies with knowledge of charging infrastructure. According to one of the industry actors, there is a lack of seeing the full picture and having a system mindset.

One representative of the policy makers also explained that the development of the electric aircraft is fragmented and is in need of clear goals that the actors can move toward. The representative further stated that it is important to establish that this innovation is a priority. More initiatives need to be taken by the government. The innovation still faces inherent inertia and resistance from some incumbent actors that do not fully believe in electric aviation. They also find that simply discussing the matter within national borders will not be enough. They desire more cooperation and collaboration on an international level.

Table 14: The actors' opinions of their collaboration.

Industry	• *"We have had some contacts with Avinor, which is in Norway and operates the Norwegian airports, and that we also try to learn from each other and see what has been done in different places." – SRF, March 2021* • *"To get all the players in the Nordics to fly and talk together. At first we did not want to include the electricity companies, but now it is good to have them with us. To include Tesla for superchargers, as it can fit electric flights. It is new value chains that are created and these are experiences that we do not have and then all the skills are needed on site. There is an advantage in having a systems mindset where you see all the pieces at the same time." – NEA, April 2021*
Researchers	• *"If I compare the road vehicle industry with aviation, the aviation industry is completely different – both nationally and internationally, it is connected in a way that you need the change to succeed. You basically need to change everywhere. It is not enough that we in Sweden invent our own solutions, but that there is a dialogue and an exchange of knowledge with the rest of the world and of course in the other direction." – RISE, February 2021* • *"There is this network, NEA. It gathers all the actors in this sector. From airports to airlines, agencies and developers, us from the institutes, academy and so on. To establish common goals and move together in order to reach this." – RISE, February 2021* • *"There is a lot of enthusiasm and then there is some inertia from the established, both researchers and industry partners who are a bit reluctant and do not really want to drive forward the development at the same pace as us." – ELISE, March 2021*
Policy Makers	• *"But I think the picture is still fragmented due to many actors and different agendas. It may be that some should be climate-smart and others business-competitive. You want the market to do its own thing and find its own solutions, but then a lot of resources are spent on uncertain things. It is about putting resources into the right things, so we can end up where we want in time." – Trafikverket, March 2021* • *"There is a willingness to cooperate [...] to tie everything together and get the whole thing to work [...]. In terms of airports, charging infrastructure, manufacturers, and airlines, etc. for development. There is pretty good cooperation there." – Trafikanalys, March 2021* • *"The cooperation is good, but it could be improved upon." – Trafikverket, March 2021*

4.6.2 Timeline

All interviewees speak of the same general timeline regarding the development of the different aviation solutions, illustrated by *Figure 7* and composed in *Table 15* as well as *Table 16*. The consensus is that the biofuels solution, which does not demand any changes on the technological side, is already present in aircraft. The opinions are scattered around how fitting of a solution it is. Some see the biofuels as a temporary solution that will be obsolete as soon as other alternatives mature and others see it as a potential permanent solution.

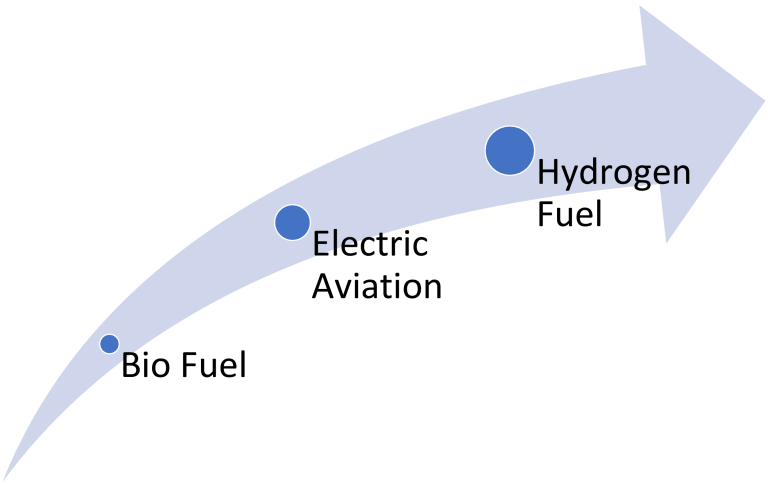


Figure 7: Chronological introduction of the different innovations.

Secondly, battery-powered electric aviation will be released in the upcoming years. The current dates for the release of the first-generation Swedish prototype with full certification are around the year 2026, based on the ambition of Heart Aerospace. The serial production of the aircraft for national commercial use is predicted to be ready in 2030. Most actors share the opinion that these will be ready by then. Others do find it somewhat ambitious. One of the researchers stated that Heart Aerospace is not the only company that is developing electric aircraft; there are several other projects as well in the pipeline to be released. Regardless of Heart Aerospace's success, a larger electric aircraft will be produced by 2030, at the latest.

Many companies have recently begun to invest in the hydrogen solution. It is the newest alternative for Sweden so far and has gained some traction during the recent year. Research in Sweden has still not made much progress with this option. There are a multitude of problems that need to be solved, such as the insufficient production of green hydrogen and issues with its storage.

Table 15: Timeline for electric aviation

Industry	• *"There is a company called Heart Aerospace that is developing an electric aircraft [...] The ambition is to have a 19-seater aircraft that will fly commercially in 2026. This would then mark the first opportunity with commercial flights using electric aircraft in Sweden."* – *Swedavia, March 2021* • *"They talked about 2025 first but now it is 2026. It may be until 2030 for more aircraft, but somewhere between 2026-2030."* – *NEA, April 2021* • *"When will this aircraft start flying? Maybe 2026 [...] Or they will come in 2028."* – *SRF, March 2021*
Researchers	• *"Between 2025 and 2030, then we will probably see an introduction of electric aircraft. There is a lot going on, it is not just Heart Aerospace, there are quite a few other projects."* – *Heart Aerospace consultant, March 2021* • *"I can further report what they say publicly and that is they have certified 19 seat aircraft in 2026. It is a pretty realistic plan I would say, within the category they are aiming for. It should be possible in principle to certify an aircraft in three years from when they have a good enough prototype."* – *ELISE, March 2021* • *"There is the government plan that I think is starting to come into force by this summer, where you want to have reduction obligations to see an increase in the mix of biofuels in the future, 2030 or something like that. [...] that during this period you start working on more fuels, new vehicles, electric aircraft and things like that will pop up."* – *RISE, February 2021*
Policy Makers	• *"Heart Aerospace said that they will launch it in 2025, now it will be 2026. Rolls Royce then came out with the idea that they would build a larger one by 2025. There are initiatives that pop up a bit here and there, but it is sluggish in the beginning. Should there be a market for Heart Aerospace aircraft and it is launched in 2025, there will be an explosion of more electric aircraft. Then more people will jump on this. We are starting to talk about things starting to happen by 2030, because there must be a majority of flights in the air for it to be an established market. But it is pretty early, 2030 I think."* – *Trafikverket, March 2021* • *"2026 is the year that Heart Aerospace has set as the target for its passenger flights on electric power."* – *Transportföretagen, March 2021* • *"There is, as you know, Heart Aerospace which has the ambition to have the plane up in 2026. They have, due to the pandemic, postponed the target to 2026 from 2025."* – *Trafikanalys, March 2021*

Table 16: Time of introduction for the different alternative fuels.

Industry	• *"We have already used SAF (sustainable aviation fuel) since 2011. So this is currently in use and has the potential to save up to at least 75% CO_2 emissions compared to fossil fuels." – KLM Royal Dutch Airlines, April 2021* • *"So right now, it is really biofuel, because that is what you can refuel in conventional aircraft and biofuel is there – even though there is too little." – SRF, March 2021* • *"And then when we will have hydrogen, which may actually be a better option than battery-powered aircraft. But it is further in time." – SRF, March 2021*
Researchers	• *"A low-hanging fruit is of course different types of biofuels. We already mix a small amount of biofuel in the jet fuel that aviation uses." – RISE, February 2021* • *"Then there is also flying on biofuel. When it comes to removing liquid fossil fuels, it has the greatest effect on emissions. Because you can use biofuels in today's aircraft, which you can already replace up to 50 percent today." – Heart Aerospace consultant, March 2021* • *"It requires brand new planes with a brand-new design to bring along hydrogen. It also requires a proper distribution around the hydrogen gas itself and hydrogen production and much more. It has a very large potential because it has a high energy density. Sure, it is very exciting, but not even those who develop it are aiming for an introduction before 2030." – Heart Aerospace consultant, March 2021* • *"The disadvantage of hydrogen is that it has technical problems [...] there is some initiative from the big players, such as Airbus. [...] They said that they will have hydrogen aircraft in operation by 2035." – ELISE, March 2021*
Policy Makers	• *"Biofuels are a temporary solution and this will not make us reach climate neutrality by 2030 but only as close as possible. That is why we have all other solutions such as hydrogen and electric aviation." – Trafikverket, March 2021* • *"I think these battery-powered electric planes will come first. They have come a long way. When it comes to hydrogen, a lot is invested in it, because just a year ago there was no desire for it. It has now arrived very quickly." – Trafikanalys, March 2021*

5 Analysis

This analysis starts off by incorporating the MLP part of the theoretical framework. The first step involves determining and mapping out all elements of the transition. This can be found in the first two subsections. The subsequent two subsections introduce a new concept that makes overcoming all relevant barriers easier. The last subsection incorporates dominant design theory.

5.1 The Elements of the Transition

According to Geels (2011), MLP views each analytical level in transitions as heterogeneous configurations of elements. The aviation industry can be conceptualized in the same manner. One important actor group in the aviation industry is the airlines. They enable commercial travel through airspace. They have, so far, been operating and maintaining conventional technologies, such as the jet-engine operated aircraft. This has been one of the crucial elements in the socio-technical system, until now. According to their business models, it is of interest to continue to use large aircraft capable of transporting many people over large distances. This enables economies of scale. This is due to more passengers sharing the costs of the pilot and fuel as well as larger combustion engines being more efficient than smaller ones. Whenever the lifespan of these aircraft runs out, the actors repurchase similarly sized models to continue the business activities. This has been the most efficient way to maximize the profit in their business models. For this reason, the incumbent airlines originally did not have any strong interest in or any economic incentives for changing the conventional fossil-fueled technology for the newly emerging niche innovations. Before this ongoing transition, the current business opportunities had greater value.

The old technological elements have, however, been faced with pressure from the socio-technical landscape. This stems both from the rising concern for sustainability in the public opinion and political pressure to reduce climate impact. Due to changes in the public opinion, new cultural phenomena such as *flight-shame* have appeared (Söderberg & Wormbs, 2019). This important element being the customer practice, has accordingly changed in favor of more sustainable options to flights, such as trains or buses. The authorities have also set ambitious goals for sustainability for 2045 (Karlsson et al., 2020). As a result, new more sustainable innovations have appeared. Alternative fuels include full electric battery-powered aviation, hydrogen-powered aviation, and biofuels.

One of the leading projects of the new developing sustainable technology in Sweden, is the ES-19 model from Heart Aerospace. The change in technology is supported by the updated scientific knowledge from research institutes. This scientific knowledge is a central element in the transition. Examples of such research institutes and projects include RISE and ELISE. The research includes the feasibility of related elements to electric aircraft, such as standardized charging infrastructure. Other important infrastructural elements include a sufficient energy supply.

Policies and flight regulations are other important elements. These were created with the traditional jet-engine operated aircraft in mind. These have yet to be adapted to a sustainable mindset. An example is the *EU-aviation enactment*, which does not support alternative fuel options. This is partially due to electric aircraft still being developed and not having reached maturity. Another element affected by the nascent state of electric aircraft is the certification, which cannot be completed before the technology is ready. An introduction of an electric aircraft will demand full examination

and certification from the Swedish transport agency Transportstyrelsen guided by EASA. The last important element is the cultural meaning. This is how the public perceives aviation – what practices they undergo, such as waiting at the gate for hours before stepping aboard the aircraft and how often they use it. Other present examples include going through passport-checks and various security measures. Aviation is used to travel distances, where other means of transportation are inconvenient or impossible.

5.2 Electric Aviation in Sweden

The socio-technical approach highlights multi-dimensional interactions between the elements of a socio-technical system. The MLP distinguishes this multi-dimensionality and maps out the elements into three analytical levels of niche innovations, socio-technical regime and socio-technical landscape (Rip & Kemp, 1998; Geels, 2012). An illustration of the electric aircraft development together with related elements can be observed in *Figure 8*.

The niche-innovation level consists of technological innovations in smaller environments. These niches act as limited environments for nascent innovations and development. These can either take the form of market niches or technological niches (Geels, 2006). The electric aircraft is still a very new concept and aircraft for commercial travel usage are not fully developed yet. These are still being tested in controlled environments, i.e., technological niches, such as in the start-up company Heart Aerospace in Sweden. However, examples of other electric aircraft are available. In Sweden between Sälen and Trollhättan, there are regular flights with Pipistrel's Taurus Electro G2, which is a two-seated electric aircraft – available for private usage and educational purposes. This can be identified as a market niche. This is due to the fact that it cannot be used for commercial passenger commute, due to its lacking passenger capacity. There is, strictly speaking, no aircraft that are viable for extensive commercial flights in Sweden.

Relevant technological niches also include projects of sustainability transitions within the regional airports of Sweden. The socio-technical regime level actor SRF has begun a transition into having fully sustainable airports through the project called *Grön flygplats*. This project has begun with transitioning Örnsköldsvik airport. Together with Vattenfall, the electricity producer, multiple aspects of the airport such as the charging of vehicles, nearby transportation means, and power generation are going to be updated to be more sustainable. Supporting the development of electric aircraft are project ELISE and the projects of the research institute RISE. ELISE is accelerating the progress of the development of electric aviation in Sweden. It involves technological niches on the holistic scale, how everything fits together. Aspects of this are electric engines with high efficiency and low mass. RISE is also involved in many aspects of the development. One such aspect is the charging of vehicles and placement of the charging infrastructure with regards to how the vehicles move. The niche environment also provides space for the creation of networks that promote niche innovations, consisting of actors on both niche-innovation and socio-technical regime levels. The NEA and Transportföretagen provide support for the actors developing electric aviation. The aim is to establish common goals and concrete collaboration.

The socio-technical regime-level is made up by cognitive routines and activities of social groups that reproduce and maintain the elements and linkages in socio-technical systems (Geels, 2006). This refers to the actors involved in the alignment of existing technologies, policies, regulations and

innovation infrastructure. Airports, airlines, scientists, policy makers and societal groups are all examples of this. Both regional and state-owned airports, i.e., SRF and Swedavia, are two important actors on the socio-technical regime level. They are both involved in technical niche projects to have the airports adapt to the new innovation. Institutes like RISE and incumbent airlines, such as Finnair and KLM are all part of the socio-technical regime level. Lastly, policy-makers include agencies such as Trafikanalys, Transportstyrelsen, and Trafikverket. Trafikanalys investigates different arrangements and measures in the transportation sector under the mission of governmental agencies. Transportstyrelsen is responsible for regulations in road, water and airspace. Trafikverket is responsible for the long-term planning, building and maintenance of state-owned roads and railroads.

The last analytical level of MLP – the socio-technical landscape, covers the wider context. It consists of cultural or normative values as well as material arrangement (Geels, 2006). The material arrangement refers to the roads, spatial arrangements of cities and electricity infrastructures. The actor enabling the electricity infrastructure is Vattenfall. On the conceptual side there are the public opinion, trends and normative values. One growing trend is the idea of sustainability. People are conscious of CO_2 emissions and want to reduce their climate footprint. An example of this is the newly emerged phenomena *flight-shame*, where other forms of transport are taken instead of aviation. These recent changes have created pressure on the existing socio-technical regime. Together with the momentum that niche-innovations, such as the ES-19 has, this has caused destabilization on the socio-technical regime level. The regime elements no longer align with the other two levels. One important actor on the landscape level is the authorities. It is from such actors that landscape pressure originates. In the present, their impact includes laws, clear goals for sustainability and policies.

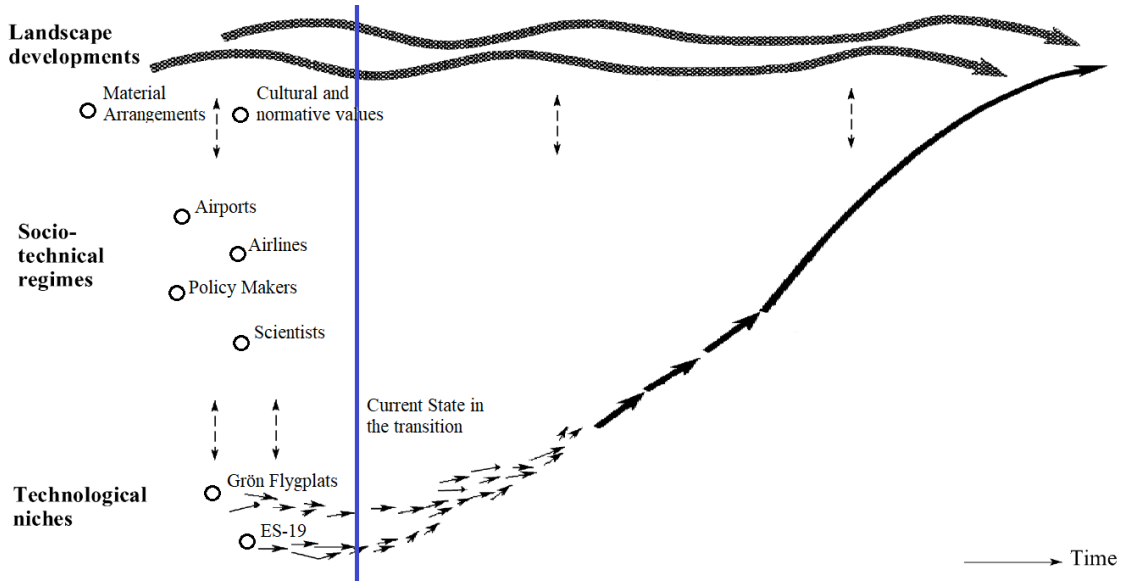


Figure 8: A simplification of the electric aircraft development and its position in time. Inspired by the work of Geels (2002).

5.3 *Socio-Technical Gates* – The Interplay of Actors

The findings show that there are multiple barriers that hinder the development of the electric aircraft. Regarding the financial condition, there is a lack of funding for the present research as well as lack of funding for infrastructural preparation on the airport side. The perceived technological barriers are insufficient battery density, no set standard for charging infrastructure, and the assembly of the electric aircraft. The certification procedure is perceived to take a long time, which would be a barrier as well. The regulatory conditions for the airports are also considered a barrier, as the airports do not know what to expect from the technology. The lack of political instruments and clear goals was also perceived as a barrier. Lastly, not all actors that are involved in the transition believe in the innovation.

Geels (2006) argued that transitions come about when the actor dynamics at different levels align and link up. There were multiple barriers for the transition of the jet-engine mentioned in that article. Instead of looking at individual barriers, the actor dynamics were instead highlighted. Transitions cannot be made with a breakthrough in a single aspect. Linkages between co-evolutionary dynamics is what enables the transitions. Key aspects of the multi-level dynamics lie in the communication and collaboration capabilities. Although Geels's empirical case did not treat the landscape level, he claims that the interaction between the different regime-elements and niche-innovation technologies was clear. This means that if there is something that makes overcoming the barriers between the niche-innovation and socio-technical regime levels easier, it should exist in these linkages.

In this empirical case, most of the interviewees perceived the communication and collaboration between them and other actors to be good. The findings, however, show that their focus was on the nearest collaboration with other actors regarding their own projects. Actor collaboration between them and other projects was seemingly non-existent. The collaboration did also not extend far vertically between the analytical levels. The researchers and developers mostly collaborated with similar actors in the niche-innovation level. Policy makers focused on the elements and actors that exist on the socio-technical regime and landscape levels (such as airports, airlines and governmental organs).

There is an overlooked aspect of the transition that acts as linkage between most niche-innovation and regime barriers. This aspect is more important than overcoming individual barriers. In the case of Geels (2006), this aspect corresponded to the actor dynamics in general. In this case, however, it is more specifically the communication and collaboration between the actors relative to the electric aircraft. In this thesis, to separate this special aspect of communication and collaboration from other elements and barriers, a new concept that encapsulated this aspect is introduced – the *"socio-technical gate"*. Metaphorically passing through this *socio-technical gate* would align the elements of the niche-innovation and socio-technical regime. This would in turn enable overcoming the barriers in an easier manner. An illustration of this can be observed in *Figure 9*.

5.4 *Socio-Technical Gates* – The Legitimacy of Electric Aircraft

The alignment of the elements on the niche-innovation and socio-technical regime level can be completed by passing through the *socio-technical gate* as specified in section 5.3. However, it is important to align the elements on all three analytical levels of the MLP and not just the two first levels. The remaining elements are on the socio-technical landscape and its connection with the

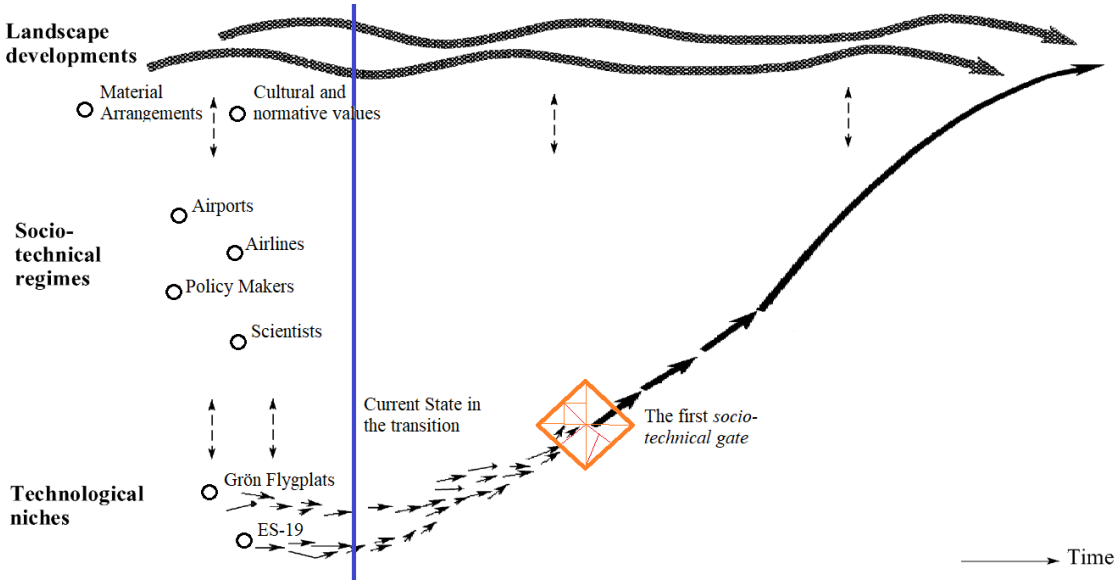


Figure 9: A conceptualization of the first socio-technical gate.

other two levels.

There exist barriers between the landscape and the other socio-technical levels. On the financial side, there is a lack of funds for future research. The current business models for conventional aircraft are not suitable for electric aircraft. For the regulatory conditions, safety measures need to be specially implemented for electric aviation. This includes the risks of high voltage and dangerous battery fires. Airline policies might also need to be updated. Finally, for the political conditions, no clear goals have been set for electric aviation. There is also a lack of political instruments to ease the introduction. The fact that there existed a *socio-technical gate* that linked the elements and barriers between the niche innovation and socio-technical regime levels, implies the existence of a second *socio-technical gate*. This would then be positioned between the landscape and the other socio-technical levels.

Geels (2006) explained that the jet engine was slow to enter American civilian aviation, due to various uncertainties. Airline companies were hesitant over fuel costs, the high temperature the engine operated at and stress on the technological elements. There were also mismatches with the length of the runways. The empirical findings in this thesis show similar results. Not many actors believe in the technology so far. There have been doubts about the battery capacities and the size of the aircraft. There is a clear issue with the public faith in the innovation, i.e., a legitimacy problem. Further supporting this theory, is that the factor that was instrumental in overcoming the last barriers for the jet-engine case, was showing that it worked. After the Comet was presented to the world as technically feasible, the public loved it. Airlines scrambled to purchase the aircraft. The American transportation organ, the Federal Aviation Administration, was put in charge of updating the safety measures for the jet-engine (Geels, 2006). In a similar manner, there is a high chance that proving that the electric aircraft is technically feasible, would lead to it solving the legitimacy

aspect. This would in turn make relevant barriers between the socio-technical regime and landscape levels easier to overcome. This cements the legitimacy aspect as the second *socio-technical gate*, as illustrated by *Figure 10*.

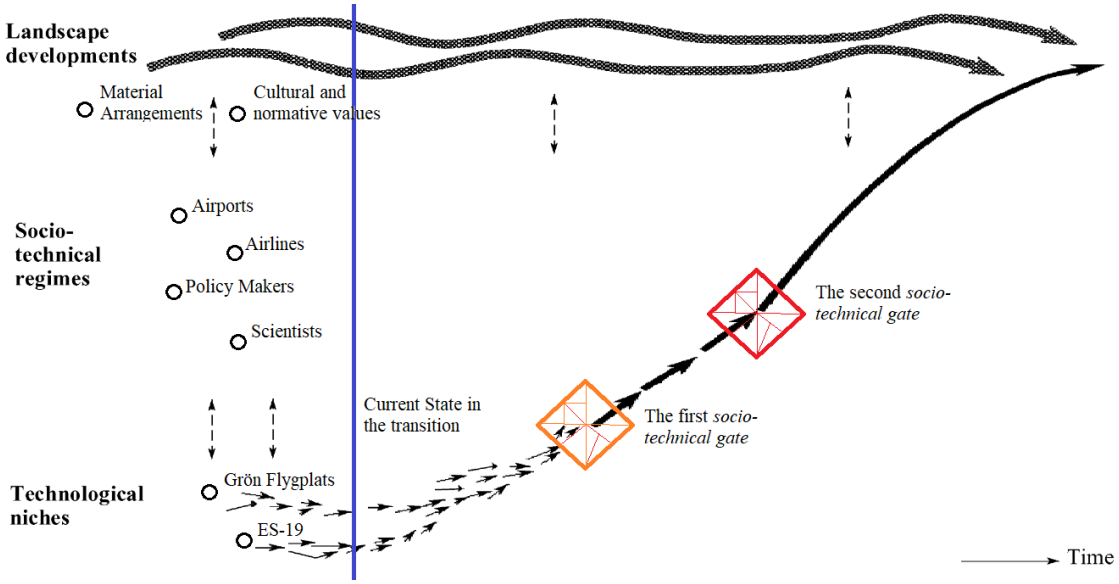


Figure 10: A conceptualization of the second socio-technical gate.

5.5 The Process of Achieving Dominant Design

In dominant design theory, there are primarily three important factors: timing, window of opportunity and dominant category. Most research has moved on from timing and the subsequent attributes, such as the first-mover advantage, and instead focuses on the phenomenon of the window of opportunity, which commences at the convergence of the dominant category. The window of opportunity is treated as an exogenous event. When it commences, multiple companies who see the worth of the new innovation would design and launch their own versions of the product (Lee & Malerba, 2017). An example of this would be the introduction of the Comet in 1952. After it was proved that it was functional, different companies scrambled to create their own versions. Boeing created the civilian Boeing 707 and Douglas Aircraft had made the jet-tanker, Dash-80 (Geels, 2006). The introduction of the Comet marked the convergence of the dominant category "jetliners". At the end of the window of opportunity, the winner was crowned as the dominant design. In the turbo-jet example, the Boeing 707 became the winner (Geels, 2006). Another example is the minivan industry. In its early phases, the different categories were known as "minivan", "compact van", and "people mover". After the convergence of the definitions, this eventually led to the emergence of a dominant design (Suarez et al., 2015).

According to most literature in this field, the window of opportunity is spontaneous and is solely an exogenous event. However, according to Yap and Truffer (2019), the window of opportunity can become endogenous. This process is made possible through the interplay of different actors, trying to shape different layers of the selection environment. By doing so, the actors can collectively determine the dominant category, which marks the start of the window of opportunity. An example of such interplay in this transition could be to develop charging infrastructure on the niche-innovation level, developing relevant policies and regulations for this on the socio-technical regime level, and

preparing the physical location for such infrastructure on the landscape level. All such interplay can be made easier if all elements are aligned. Passing through both *socio-technical gates* would do so. According to Geels (2006), it is possible to coordinate the interplay and actor collaboration through building socio-technical networks. Using the turbojet example, a socio-technical network was formed at the time. The goal was to align and co-construct different elements, such as sponsors and finance, technical elements, and knowledge. They included domains such as policy makers, business, law, and science. Following this reasoning in this case, creating a socio-technical network of all relevant actors, could be part of the means of passing through the first *socio-technical gate* and forming the dominant category. This would open the window of opportunity and subsequently end with the emergence of the dominant design (Anderson & Tushman, 1990).

6 Discussion

This section is divided into four parts. The first subsection discusses the current development of electric aviation in Sweden and other alternatives of sustainable aviation. The various barriers for the development are also mentioned. The second and third subsection cover the aspects of the socio-technical gates and suggested means of passing through the gates. Lastly, the final subsection covers the process of reaching a dominant design and how this can be affected by the actors.

6.1 The Current Development of Electric Aviation

Currently, electric aviation in Sweden is developing on a niche-innovation level. The significance of the innovation has recently been enhanced due to impacts from a changing landscape. The change is caused by recently established sustainability policies targeting the Swedish transportations sector to lower the climate impact to zero. There are other alternative fuels as well. Biofuels, for instance, have been offered as an option when it comes to current conventional flights, since it can already be mixed with the conventional jet fuel for a slightly lower efficiency. Biofuel can, however, only be seen as a temporary solution for sustainability, according to multiple interviewees. There is no stable supply of green biofuels and it is currently being debated whether it truly is a suitable solution or not (Hari, Yaakob, & Binitha, 2015; Tempels & Van den Belt, 2016). Therefore, other alternative fuels, such as the full electric battery-powered aircraft or hydrogen-powered aircraft have emerged. Although the hydrogen option is expected to have a greater flight range, it is also far from ready. One budding development of electric aircraft is ES-19 by Heart Aerospace in Sweden. This is the leading electric aviation project in Sweden. Several incumbent airlines, such as Finnair, have expressed interest in the electric aviation solution in the form of non-binding agreements and letters of intent for the purchase of such aircraft. There are also other projects under development in other countries. The aim is to have the ES-19 fully operational and certified for commercial flight within five years from now. The year 2026 has been set as the specific year for its introduction. Moreover, there is collaboration between the actors within the industry to ease the transition to electric aviation.

The interviews show that all actors believe that there are a multitude of barriers to the development of electric aviation. The barriers reside in every analytical level of the transition and are not only connected directly to the aircraft itself. These include, the complexity of aircraft assembling and the still nascent battery technology, lack of funding for technical development and infrastructure at the airports, the need for a certification, lack of suitable regulations and political support, issues of legitimacy, and the need for new business models. In more detail, the complexity of assembling the aircraft and integrating the parts was perceived to be a problem. The interviews showed that several actors – even those believing in the technology – considered the current battery technology to be a barrier to the development of electric aircraft. They perceived the continuous development of the battery density and efficiency as a necessity for the development of electric aircraft. Lack of funding was a major problem in all aspects of the development. Both the preparation of the infrastructure and the development itself require more investments. The developers are not capable of financing this privately. The certification process could also be a significant barrier, should it take years to complete. However, in the case of the ES-19 model, the process has already begun. The regulatory aspects are also seen as barriers. Authorities need to define new airspace regulations, as the old ones might not fit the electric aircraft. The airlines and airports might also need to address different regulatory aspects, such as security measures. For instance, how far away from ordinary

aircraft infrastructure, electric aviation can be placed and other safety measures, such as what they will need to do if a fire breaks out. Lastly, the last problem was an issue of legitimacy. Not enough people, including the authorities, believe in the innovation. There were some concerns about a need for new business models for the airlines as well. The current business models of the incumbent airlines are not compatible with the new innovation.

6.2 The *Socio-Technical Gate* of Actor Collaboration

According to the analysis, passing through the first *socio-technical gate* would make overcoming related barriers easier. This is a matter of communication and collaboration between the actors on the niche-innovation and socio-technical regime levels. With better collaboration, the research projects could perhaps apply for grants using similar means like the airports did. SRF did receive grants from the European Regional Development Fund for instance. With better communication, the actors who perceived the battery technology to be insufficient, would understand that it was a misconception. The batteries are fully capable of flights up to 400 kms which is suitable for shorter inter-regional flights. This is possible by using the right aerodynamic trajectories. Similarly, Heart Aerospace is making great progress with the assembly of the aircraft and the certification process, these two processes being held parallel to each other. This is made possible by certifying combinations of parts at a time, rather than trying to certify a complete aircraft. Furthermore, if the airports would communicate with experts in the charging infrastructure development, they would also better understand what is necessary for preparing the infrastructure. With better communication, the actors involved should trust the technology more. Lastly, while political conditions are hard to affect, collectively deciding on larger goals might have a greater effect than smaller more individual goals. While it is possible to overcome individual barriers to the development of electric aviation, the linkage between the barriers makes it easier to first overcome the actor collaboration problem.

Seeing the benefits of passing through the *socio-technical gate*, the next step would be to identify possible means of passing through the gate. Geels (2006) identified that forming socio-technical networks for the actors would strengthen the interplay between the actors and solve the actor dynamics. These networks are a plausible means. In this case, for the socio-technical networks, both Transportföretagen and NEA are trying to connect the relevant actors. NEA is for instance a socio-technical network project. It is formed mostly by actors that affect the transition directly, such as the developers, airlines and other aviation companies. They have, however, limited the contact with actors affecting the transition outside of the aviation industry. Examples of this are electricity producers, e.g., Vattenfall, as well as battery and charging experts, e.g., Tesla or Northvolt. The representative of NEA admitted that its members originally did not consider admitting such actors into the network, but that their importance had now been reassessed. Communication and collaboration with other countries could also be improved. The actors of other countries involved in electric aviation, such as Norway, could perhaps also be included.

6.3 The *Socio-Technical Gate* of Legitimacy

While socio-technical networks are a plausible means for passing through the first *socio-technical gate*, it does not overcome the remaining barriers between the landscape and other levels. There is a need to similarly pass through the second *socio-technical gate*. After passing through the *socio-technical gate* of legitimacy, the remaining barriers will be easier to overcome. One example is the financial

conditions. Funding for future research can perhaps be obtained if investors and authorities believe in the innovation. The current business models for conventional aircraft are not suitable for electric aircraft. If the public believes in the innovation, then airlines will realize that there is a demand for it. The airlines will then figure out better and more suitable business models. Similarly for the regulatory conditions, after the airline companies believe in the innovation, they can communicate with developers to set regulations for the aircraft. The biggest change would, however, be on the political side. When authorities see that it is functional and has potential, they would set clear goals. This could for instance be to replace all inter-regional airplanes with electric aircraft. The authorities can additionally, invest in further development of the technology and set up incentives to use the electric aircraft. An example of this, as stated in section 4.1.2., could be loan guarantees for buying the aircraft and discounts for operating the technology. Such political instruments and measures were also suggested by Trafikanalys to the authorities.

According to multiple interviewees, as long as one single electric aircraft is showcased or shown to be operational – that it can handle a flight, then this would show that the technology works and that it has market potential. Supporting this claim is the jet-engine case from Geels (2006). As explained in the analysis in section 5.4., similarly to this situation, there were also uncertainties about the jet engines at the time. It was not until the emergence of the Comet, that the worries from the airlines were alleviated. As long as the ES-19 can be shown to function and have commercial potential, the public opinion will be positive. This is a means to pass through the second *socio-technical gate*.

6.4 The Process of Reaching a Dominant Design

Electric aviation will most likely reach a dominant design in the future if the two *socio-technical gates* are passed. The process up until then, however, is yet to be decided. As previously explained, most literature has perceived the window of opportunity solely as an exogenous event. In such literature, it is common to see that the window of opportunity depends on coincidences and luck. It is not certain what causes the window of opportunity to appear, it is all dependent on external factors. Yap and Truffer (2019) instead stated that this window of opportunity can be controlled internally through the interplay of different actors, i.e., creating an endogenous process. This is unique in the sense that not all industries have the capabilities to do so. In the car industry for instance, a single actor can use its competence and resources to affect the whole market by itself. Other companies become more competitive than cooperative. This is not the case for the aviation industry. As the industry is currently in decline, the actors are more closely knit than in other industries. The actors have the capability of collaborating in order to control the window of opportunity, which starts with the convergence of the dominant category.

The dominant design will not be achieved in the sense that all airplanes would become electric. The current maximum range is the limiting factor there. The battery technology is only sufficient for shorter distances. However, it also provides opportunities for the future. In the present, most airlines aim to only buy large aircraft. This is due to the fact that large aircraft bring economies of scale. By maximizing the number of passengers and distance, the overall costs are reduced significantly. For shorter distances, however, this means that the airlines also have to use these larger conventional aircraft, which costs more to use on such distances. The other option is to specifically purchase small jet aircraft that are not compatible with their business models. These smaller aircraft do not bring any economies of scale. Either way, the aircraft are not optimized for such travel. Through

the empirical data, it can be confirmed that the biggest opportunity lies in these short distance flights for electric aviation. The business model barrier mentioned earlier by the interviewees is actually more of an opportunity. These electric aircraft are much more cost-efficient compared to conventional combustion aircraft. One of the researchers stated for instance that one electric engine cost hundreds of thousands SEK, while a jet-engine cost several million SEK. By only using the electric aircraft on these shorter ranges, there are also business model opportunities. Several actors speak of airborne bus traffic. By setting up a network of closer flight hubs, you could also travel further, using more "bus" stops. Policy makers also envision this to take over smaller routes where larger aircraft are less efficient and routes where there are no other options, such as over water. This also includes inter-regional flights. Electric aviation might then be competing with other parallel socio-technical regimes, such as bus or train transportations. Since large incumbents cannot simply discard older aircraft as these can still be used, this gives newcomers a unique opportunity to enter the market. These smaller actors might need financial aid from the authorities, however. Passing through the *socio-technical gate* of legitimacy as discussed earlier might solve this as well.

It is important to note that there are other companies that are also developing full electric battery-powered aircraft, such as the nine-passenger seated *Eviation Alice* (Baumeister et al., 2020). Such projects are also expected to reach maturity within 2024-2030. Interest has been shown in both Heart Aerospace's solution as well as *Eviation Alice*. The first project to reach maturity will likely mark the start of the window of opportunity. The concept of electric aviation would be defined, i.e., the convergence of the dominant category. Various companies would then start creating their own electric aircraft. It is important to understand that regardless of where the dominant category converges, the dominant design will still eventually emerge from all competing alternatives and not just the first emerging option. Aligning all elements in Sweden would, however, help the progress of Swedish alternatives to the electric aircraft. One suggested way to align all elements on all three analytical levels of MLP is to pass through the two *socio-technical gates* of actor collaboration and legitimacy. Passing through these gates would likely also ease overcoming the related barriers as well. With the alignment of all actors and elements, the dominant category would converge into a single concept. Should the elements align, the concept could be that electric aviation consists of a conventional airplane body with an electric engine and powered by batteries. The timing would be decided at the actors' discretion, meaning that they can more easily prepare for the window of opportunity. At the end of this window of opportunity, one company will set the standard, which will be the standard for any future developments. This accounts for not only the appearance of the aircraft, but also the charging solutions. This industrial standard that appears at the end of the window of opportunity is the dominant design.

7 Conclusion

Presented below are the conclusions drawn from the findings and analysis in this thesis. Both research questions are properly addressed. Lastly, is an evaluation of the theoretical framework as well as possible directions for future work.

7.1 The Key Barriers for the Development of Electric Aviation

Addressing the first research question, many barriers for the development of electric aviation were identified. According to the thematic analysis performed, there are mainly four categories of barriers, these being the Financial, Regulatory & Political, Business Model & Legitimacy, and Technical. Within these themes, there are barriers such as the still nascent battery technology, lack of funding for both technical development and infrastructure at the airports, the complexity of the aircraft assembly, the need for certification, lack of suitable regulations (mostly safety measures and rules), lack of political support and clear goals, issues of legitimacy, and the need for new business models. While it is possible to address each barrier individually, this thesis has followed the ideas of Geels (2006), to use a more holistic perspective. From there, two *socio-technical gates* were identified, treating the actor collaboration and legitimacy aspects, respectively. This thesis argues that passing through these two *socio-technical gates* would make overcoming most of these barriers much easier.

For the first gate, the identified means of passing through the gate was socio-technical actor networks. Here, applicable networks include the efforts of Transportföretagen and NEA. They would have to include all the actors, even the electricity producers, to solve the problem. With the socio-technical network in place, better interplay and actor collaboration is made possible. For the second gate, the identified means of passing through the gate was to demonstrate that electric aircraft work and have market potential. According to the judgment of multiple actors, as long as a single electric aircraft – as a simple symbol – can be proved to be functional, then people will believe in it. This is also what happened in the jet-engine case (Geels, 2006).

7.2 The Emergence of the Dominant Design

The process of reaching a dominant design is complicated. Contrary to most of the extant literature so far, which sees the window of opportunity as exogenous and impossible to affect, this thesis instead argues that the window of opportunity can be made endogenous and controlled if the right conditions allow for it. These conditions are the interplay of the actors (Yap & Truffer, 2019). Accordingly, all elements of the transition must be aligned in order for this to happen. Passing through the *socio-technical gates*, identified above, will align these elements. For Sweden, this would essentially mean that the dominant category would converge at the introduction of the ES-19, if the relevant elements were aligned. The dominant category would for instance be that electric aviation consists of a conventional airplane body and an electric engine, which is powered by batteries. The timeline is also a very important factor, as there are multiple other projects also in the development of electric aviation. It is estimated that the dominant category will converge at the latest by 2030, as many of these projects aim to be completed between 2024-2030. After the convergence of the dominant category, the window of opportunity will open and lead to the dominant design.

To answer the second research question, the process of reaching a dominant design, i.e., the window of opportunity, can be affected by the actors. The actors can align the elements of the transition

by improving their communication and collaboration using and expanding upon a socio-technical network such as NEA, followed by showcasing the innovation.

7.3 Evaluating the Theoretical Framework

The thematic analysis of the data proved to be very useful. It was simple to organize the findings into suitable themes. By doing this, it was relatively easy to extract the necessary information. The theoretical framework, which consisted of parts from MLP and dominant design theory, was moderately easy to work with. While the framework was very suitable for this particular case, this is because the aviation industry had the possibility of having good actor collaboration and dynamics. This is, however, not the case for all industries. As explained by one of the interviewees, the aviation industry is unique and closely knit, unlike the car industry for instance. Applying this framework on the car industry would be implausible as the actor dynamics do not fulfill the necessary conditions. There is too much competition in that industry. The theoretical framework can, however, be used in cases with similar conditions, where great communication and collaboration between the actors is possible.

The theoretical framework used was sufficient for its intended purpose. However, it can be expanded and improved upon for future usage. Currently, it only takes elements according to MLP theory in consideration. There could perhaps be other critical elements besides these. Other theories can be added in order to get a more holistic perspective. In the jet engine case of Geels (2006), the MLP could not determine the conditions between the landscape and the other socio-technical levels. In his case, the analysis was not clear between those levels. This is another limitation in the theory. While the current analysis of the interplay of actors between the niche-innovation and socio-technical regime levels is more than enough, more theoretical literature could be applied to the interplay of actors between the landscape and other socio-technical levels.

7.4 Future work

The transition itself will only intensify after the convergence of the dominant category. According to several of the interviewees, Heart Aerospace intends to start mass producing electric aircraft by 2030. Assuming that there are no external complications, the dominant category will already have converged by this time. With the release of the ES-19, the dominant category of the concept of electric aviation will be determined. Other developers will release their products within this category and these will compete to become the dominant design. This thesis focused on how the actors could affect the process of reaching a dominant design, i.e., the window of opportunity. This mostly covered the early part of the window of opportunity – the dominant category. Further research could perhaps cover the more intrinsic details of choosing the dominant design.

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Appendix

Exemplified Interview Questions

General Questions

- *Please state your name, role and the organisation you work for.*
- *How are you involved in electric aviation?*

Technological Questions

- *How long do you think the development of the new technology will take until its release?*
- *What is your take on the electric aircraft?*

Socio-Technical Questions

- *What are some social-technical obstacles that electric aircraft face when trying to commercialize it?*
- *How do you deal or plan to deal with these obstacles?*

Business model and Economical Questions

- *What business model would be appropriate for the electric aircraft?*
- *What is the current demand for electric aircraft?*
- *What is the willingness of investors to invest in electric aircraft?*

Niche development:

- *What are some ongoing projects at this time?* Niches?
- *Could you describe the project?*

Dominant design

- *What is the most prominent type of electric aircraft? Which model?*
- *How far has it reached in terms of being commercially viable?*

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